

Global transport in soaring flights

Version: 14/09/2026

Chapter 3

Thesis §3.1

Displacement and time scales

The motion a stochastic model must reproduce

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad R_f = |\mathbf{X}_f|.$$

$$M_2^{\text{flight}}(\tau) = \frac{1}{N_f(\tau)} \sum_f \underbrace{\frac{1}{n_f(\tau)} \sum_k R_{fk}^2}_{\text{mean within flight } f}.$$

Windows stay within a retained segment. The main comparison covers $10 \leq \tau \leq 10^4$ s.

Retained and eligible populations

Archive: 156,406 paragliders and 6,094 hang gliders.

Full diagnostics: 155,085 and 6,060 eligible flights, all supported segments, on a 10-s grid.

Displacement growth, distribution shape, directional preference and memory must be confronted together.

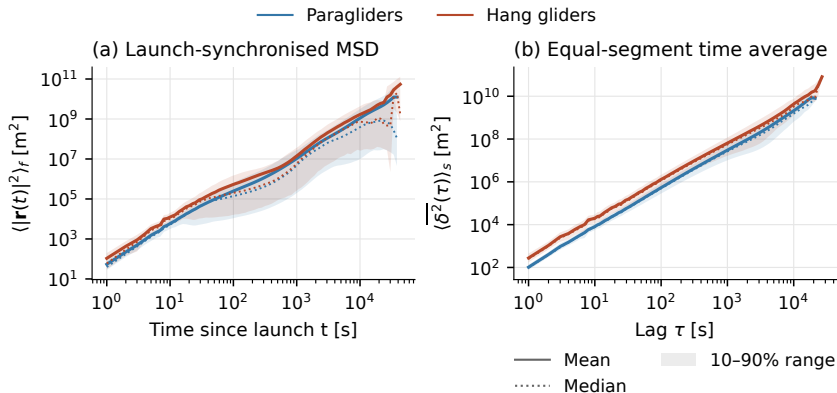
Archive TAMSD: equal segment weights. Full V_1, V_2 : equal flight weights. Baseline quantiles and moments: pooled windows.

Launch, segment and flight averages use different weights

Estimator	Unit receiving equal weight	Support requirement
Launch ensemble	available flights at elapsed time t	observed coordinate at that time
Archive TAMSD	eligible retained segments	native-grid displacement within segment
Full V_1, V_2	eligible flights at lag τ	within-segment common-grid stencil
Pooled moments	available increment origins	stated non-overlapping baseline stencil

Different exponents can reflect different weights and changing support. Their difference alone is not an ergodicity test.

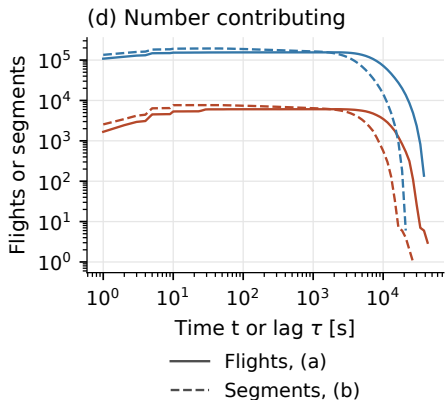
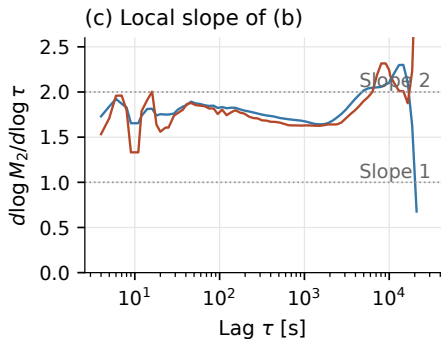
Launch averages and time averages use different populations



Thesis §3.1.1

Effective and local growth exponents

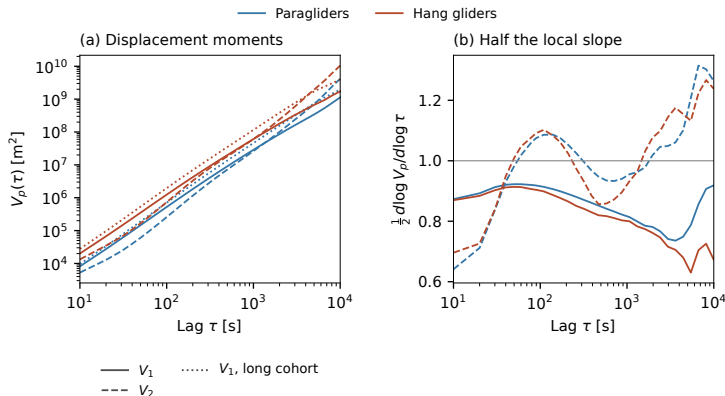
Local slopes must be read together with falling support



Thesis §3.1.2

Removing a constant velocity

The MSD grows faster than linearly on the measured interval



	b_1	b_2
Para	1.703	2.003
Hang	1.649	1.985

$V_1 = \langle R^2 \rangle$; V_2 uses second differences.

The local slope bends. One fitted number does not establish a power law.

Full eligible archive; 10–10,000 s. Solid: V_1 ; dashed: V_2 ; dotted: V_1 , fixed long cohort. Right: half the local slope. Descriptive fits; no calibrated confidence intervals.

Constant velocity cancels; an intrinsic H remains open

$$A_2 \mathbf{r} = \mathbf{r}(t + 2\tau) - 2\mathbf{r}(t + \tau) + \mathbf{r}(t), \quad V_2 = \langle |A_2 \mathbf{r}|^2 \rangle.$$

Exact cancellation

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_d t + \mathbf{Y}(t) \quad \Rightarrow \quad A_2 \mathbf{r} = A_2 \mathbf{Y}.$$

The constant velocity may combine wind, route choice and net progress. A turn or spatially varying wind survives.

At $\tau = 10^4$ s, V_2 retains only 14,360 para / 563 hang flights; it needs 2τ of continuous support.

Conditional interpretation

If centred increments have finite variance, stationary statistics and the law $K\tau^{2H}$,

$$V_2 = K(4 - 2^{2H})\tau^{2H}, \quad 0 < H < 1.$$

At $H = 1$ the prefactor vanishes. Our $b_2 \simeq 2$ cannot be read as a validated $H \simeq 1$.

Second differences measure changes of interval-mean velocity

$$\bar{\mathbf{v}}_1 = \frac{\mathbf{r}(t + \tau) - \mathbf{r}(t)}{\tau}, \quad \bar{\mathbf{v}}_2 = \frac{\mathbf{r}(t + 2\tau) - \mathbf{r}(t + \tau)}{\tau}.$$

$$\frac{A_2 \mathbf{r}}{\tau} = \bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1, \quad D_2(\tau) = \frac{\sqrt{V_2(\tau)}}{\tau} = \sqrt{\langle |\bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1|^2 \rangle}.$$

D_2 is in m/s. At $\tau = 60$ s it compares the mean velocity vector in one minute with the vector in the next minute. Changes of magnitude and direction both contribute.

Exact kinematic identity; no power law or stationarity assumption is needed.

Why a slope near two need not identify a Hurst exponent

$$V_2(\tau) = \tau^2 D_2^2(\tau), \quad b_{2,\text{loc}} = 2 + 2 \frac{d \log D_2}{d \log \tau}.$$

Saved full-archive curves

Over 10–10,000 s, D_2 varies within:

	D_2 range (m/s)
Paragliders	4.9–7.3
Hang gliders	7.6–11.5

A nonmonotone change of less than a factor two across three decades of lag.

Thesis Sec. 3.1.2; saved full-archive variations, unchanged numerical run. Ranges are descriptive, not confidence intervals.

Interpretation and limits

The τ^2 factor can dominate growth; local curvature prevents one validated H .

The fixed long-flight cohort also gives $b_2 > 2$: flight membership alone cannot explain it. Origins and weights still change.

Finite persistence can produce a crossover without nonstationarity

A stationary centred velocity with variance σ_v^2 and normalised correlation $C(s) = e^{-s/t_p}$ gives

$$V_2(\tau) = 2\sigma_v^2 t_p^2 \left[2\frac{\tau}{t_p} - 3 + 4e^{-\tau/t_p} - e^{-2\tau/t_p} \right].$$

$$\tau \ll t_p : \quad V_2 \sim \frac{4\sigma_v^2}{3t_p} \tau^3, \quad \tau \gg t_p : \quad V_2 \sim 4\sigma_v^2 t_p \tau.$$

A local slope near two occurs between these regimes despite stationary position increments.
[This is an analytical example, not a fit to the flights.](#)

In flights, turns, glide-climb transitions and changing wind survive constant-velocity cancellation. Their characteristic times, combined with changing observation support, are a plausible origin of the measured curvature.

Thesis Sec. 3.1.2 and the velocity-covariance model in Sec. 3.6; the closed formula was checked against covariance integration.

Thesis §3.1.2

Third differences and the distribution of changes

Third differences also cancel a constant acceleration

$$A_3 \mathbf{r} = \mathbf{r}(t + 3\tau) - 3\mathbf{r}(t + 2\tau) + 3\mathbf{r}(t + \tau) - \mathbf{r}(t), \quad V_3 = \langle |A_3 \mathbf{r}|^2 \rangle.$$

For $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2$ with constant $\mathbf{a}$,

$$V_2 = |\mathbf{a}|^2 \tau^4, \quad V_3 = 0.$$

Under the centred stationary-increment power law,

$$\frac{V_3}{V_2} = \frac{15 - 6 \cdot 2^{2H} + 3^{2H}}{4 - 2^{2H}}, \quad 0 < H < 1.$$

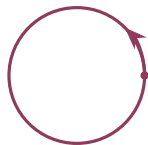
Order three requires 3τ of continuous support; the model law must hold through 3τ .

Thesis §3.1.3

Open and closed routes

Open and closed tasks change the growth of displacements

Open straight course



Closed circular course

Full eligible archive

	Task	N	b_1	b_2
Para	Open	67003	1.770	1.937
	Closed	87807	1.623	2.036
Hang	Open	2183	1.688	1.892
	Closed	3877	1.628	2.007

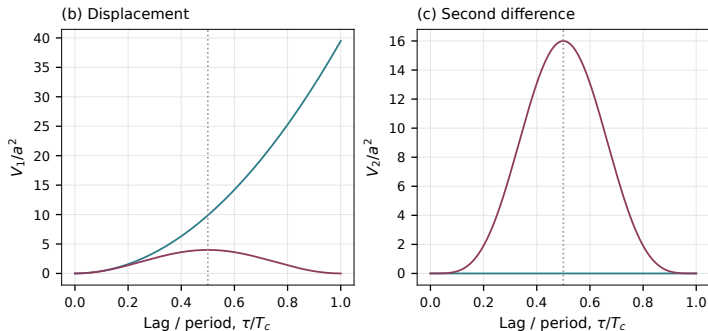
The ordering reverses at order two. Constant-drift cancellation leaves the return geometry in the observable.

Task declarations; descriptive slopes, 10–10,000 s. Drawing is a geometric illustration. [Exact circular example](#).

A circle retains its geometry after second differencing

For $\mathbf{r}(t) = a(\cos \omega t, \sin \omega t)$ and $T_c = 2\pi/\omega$,

$$V_1 = 4a^2 \sin^2(\pi\tau/T_c), \quad V_2 = 16a^2 \sin^4(\pi\tau/T_c).$$



Observed-path closure needs special care for segmented flights

$$c_f^{\text{obs}} = \frac{|\mathbf{r}_{f,\text{last}} - \mathbf{r}_{f,\text{first}}|}{\sum_{s \in \mathcal{S}_f} \sum_k |\mathbf{r}_{fs,k+1} - \mathbf{r}_{fs,k}|}.$$

The denominator omits unobserved intervals. For a segmented flight, c_f^{obs} need not be bounded by one.

Declared task, observed endpoint separation and actual flown-route closure are related descriptions with different information requirements.

Thesis §3.2.1

The scaling prediction

Scalar vector scaling also constrains east and north

With $a_\tau = (\tau/\tau_0)^H$, the vector law $\mathbf{X}(t, \tau) \stackrel{d}{=} a_\tau \mathbf{X}(t, \tau_0)$ implies

$$X_E(t, \tau) \stackrel{d}{=} a_\tau X_E(t, \tau_0), \quad X_N(t, \tau) \stackrel{d}{=} a_\tau X_N(t, \tau_0).$$

Signed marginal densities

$$p_E(x; \tau) = a_\tau^{-1} f_E(x/a_\tau),$$

$$p_N(y; \tau) = a_\tau^{-1} f_N(y/a_\tau).$$

The exponent is shared; the shapes f_E and f_N can differ. No independence or isotropy is assumed.

Magnitudes used for quantile fits

For $j \in \{E, N\}$,

$$|X_j(\tau)| \stackrel{d}{=} a_\tau |X_j(\tau_0)|,$$

$$Q_{|X_j|,p}(\tau) = a_\tau Q_{|X_j|,p}(\tau_0).$$

Positive finite quantiles allow log fits.
Negative or zero signed quantiles do not.

Thesis Sec. 3.2.1. One-dimensional densities have factor a_τ^{-1} ; the two-dimensional joint density has a_τ^{-2} .

A single displacement scale predicts equal quantile slopes

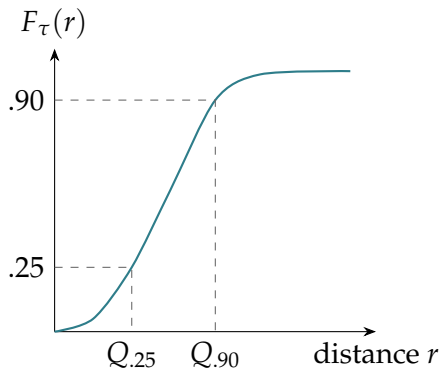
$$Q_p(\tau) = \inf\{r : F_\tau(r) \geq p\}.$$

An H -self-similar process with stationary increments predicts

$$R(\tau) \stackrel{d}{=} (\tau/\tau_0)^H R(\tau_0),$$

$$Q_p(\tau) = (\tau/\tau_0)^H Q_p(\tau_0).$$

$H_p = H$ for every p ; $Q_{.90}/Q_{.25}$ stays fixed.



Schematic CDF. Quantiles describe ranks within one distribution; they do not define permanent classes of slow and fast flights. Equal slopes are necessary, not sufficient, for process self-similarity.

From displacement windows to one quantile at each lag

For each supported origin, measure the paired displacement $\mathbf{X}(t, \tau) = \mathbf{r}(t + \tau) - \mathbf{r}(t)$, then form

$$Y_E = |X_E|, \quad Y_N = |X_N|, \quad Y_R = \sqrt{X_E^2 + X_N^2}.$$

For each quantity j , sort the values at that lag and accumulate their weights:

$$\hat{F}_{j,\tau}(y) = \sum_{f,k} w_{fk}(\tau) \mathbf{1}\{Y_{j,fk}(\tau) \leq y\}, \quad \sum_{f,k} w_{fk}(\tau) = 1.$$

Read the first observed value where cumulative weight reaches $p = .25, .50, .75, .90$. Repeat at every lag to obtain four curves $\hat{Q}_{j,p}(\tau)$.

Pooled windows: equal weight per window. **Fixed-population control:** equal total weight per flight, with common origins.

Thesis Sec. 3.2. Quantiles use inverse empirical CDFs without interpolation; flight-specific quantiles are not averaged.

Each fitted $H_{j,p}$ is the slope of one log–log quantile curve

For one discipline, quantity j and probability p , set

$$x_\ell = \log \frac{\tau_\ell}{1000 \text{ s}}, \quad y_\ell = \log \left[\hat{Q}_{j,p}(\tau_\ell) / (1 \text{ m}) \right].$$

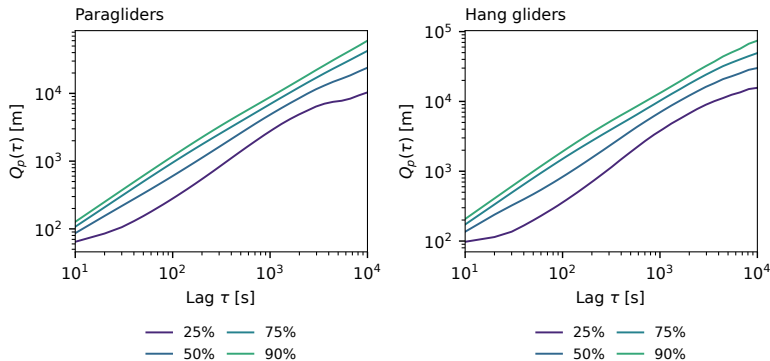
Fit a straight line with a free intercept by ordinary least squares:

$$(\hat{b}_{j,p}, \hat{H}_{j,p}) = \arg \min_{b, H} \sum_{\ell \in \mathcal{L}} (y_\ell - b - H x_\ell)^2, \quad \hat{Q}_{j,p}^{\text{fit}}(\tau) = (1 \text{ m}) e^{\hat{b}_{j,p}} \left(\frac{\tau}{1000 \text{ s}} \right)^{\hat{H}_{j,p}}.$$

- Full interval: 31 saved lags from 10 to 10,000 s, equal weight per lag.
- Repeat for three quantities and four ranks: **12 slopes per discipline**.
- A fitted slope summarises the curve; curvature can remain around the line.

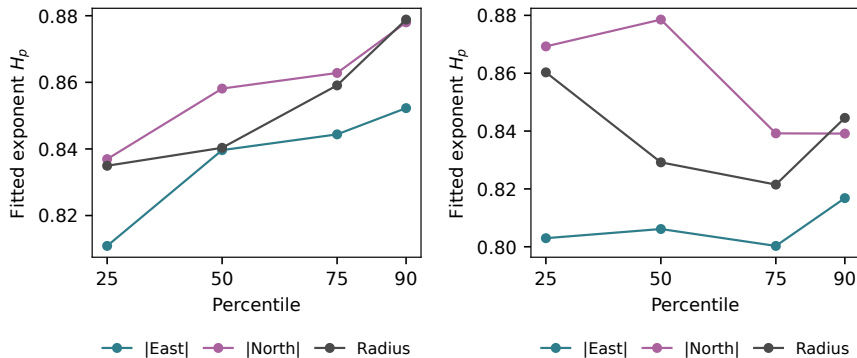
Thesis Secs. 3.2–3.3. Only positive supported quantiles enter log fits. The time reference changes the intercept, not the slope.

Archive quantile curves are the input to the fits



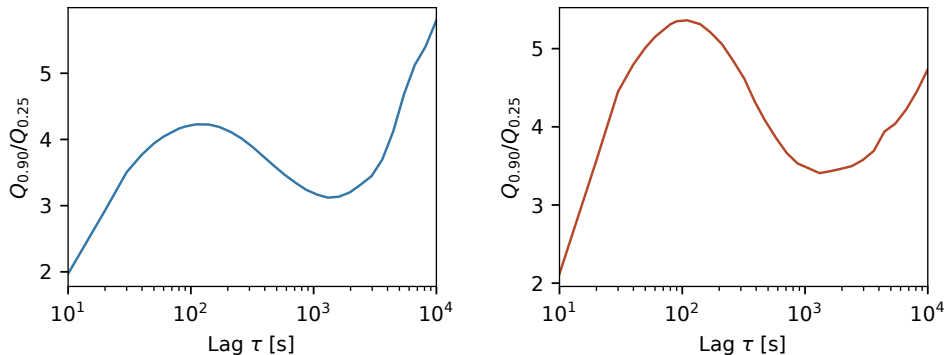
Thesis Fig. 3.4, top row. Radial quantiles; all eligible windows pooled at each lag. Purple to green: 25th, 50th, 75th, 90th percentiles.

Each point here summarises a separate quantile fit



Thesis Fig. 3.4, middle row. Left: paragliders; right: hang gliders. Fits span 10–10,000 s; the horizontal axis is percentile, not lag.

The archive quantile ratio also reveals changes with lag



Thesis Fig. 3.4, bottom row. $Q_{.90}/Q_{.25}$ uses the same pooled radial curves as the first row. Left: paragliders; right: hang gliders.

Thesis §3.2.2

A fixed population across lags

Population	Paragliders	Hang gliders
Retained cleaned flights	156,406	6,094
Eligible for the common-grid diagnostic	155,085	6,060
Fixed long-flight control	14,360	563

Every eligible segment contributes where it supports the lag. The long-flight control adds a common population and common origins across the complete lag range.

The fixed control requires common temporal support

All eligible flights contribute at each supported lag in the archive-wide analysis. The additional fixed cohort contains every flight with a retained segment lasting at least 20,000 s.

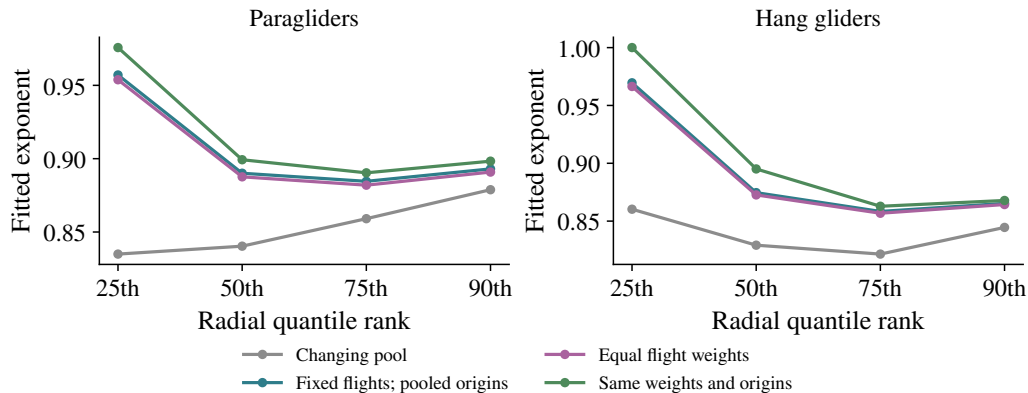
This provides at least 1001 common ten-second origins in one qualifying segment at the largest lag of 10,000 s. Shorter segments of those flights contribute wherever they support the same full lag stencil.

Holding flights fixed does not by itself hold the observations fixed

1. All available flights and origins, pooled.
2. Fixed flights, still pooling their available origins.
3. Fixed flights with equal total flight weight.
4. Fixed flights, equal flight weight and the same origins at every lag.

$$F_{\tau}(r) = \frac{1}{N_*} \sum_f \frac{1}{n_f^*} \sum_{k \in \mathcal{O}_f^*} \mathbf{1}\{R_{fk}(\tau) \leq r\}.$$

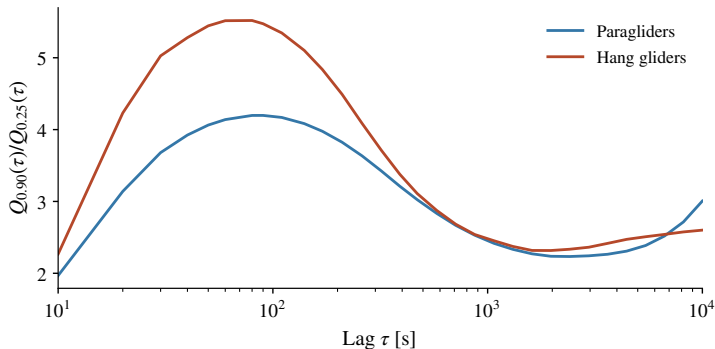
Membership, flight weights and common origins change the comparison



Thesis §3.2.3

Relative spread

Relative spread changes, but the change is not monotone



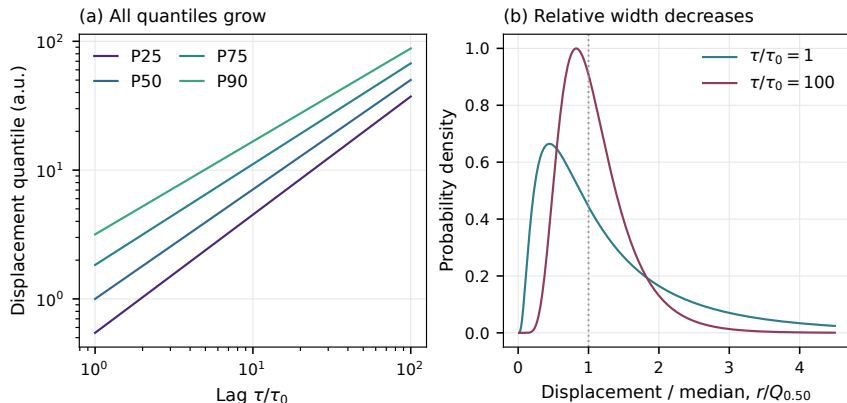
$$\frac{d \log(Q_{.90}/Q_{.25})}{d \log \tau} = H_{.90}(\tau) - H_{.25}(\tau).$$

The ratio first rises, then falls, then rises again.

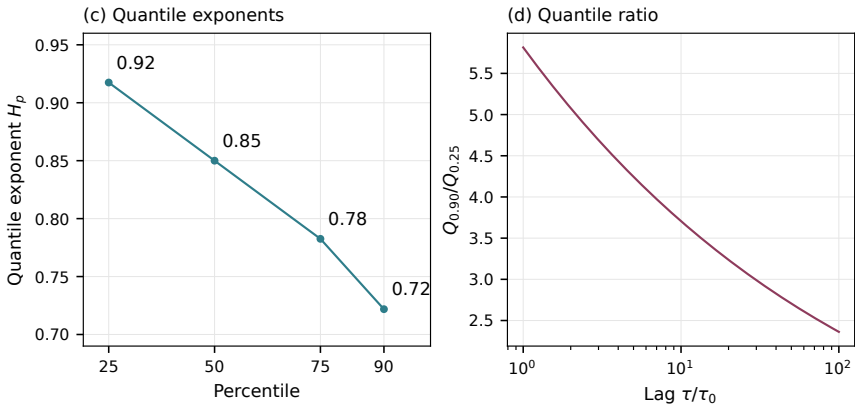
A negative fitted slope summarizes part of that evolution; it does not imply narrowing at every lag.

Fully controlled population. A lower ratio means less relative spread by this measure; it does not establish a flatter density tail.

All quantiles can grow while spread relative to the median narrows



Unequal rank exponents can encode that changing shape



Thesis §3.3.1

Testing east, north and radial scaling

Anisotropy does not prevent a common scalar rescaling

$$\mathbf{X}(\tau) \stackrel{d}{=} \tau^H \mathbf{Z} \implies |\mathbf{X}(\tau)| \stackrel{d}{=} \tau^H |\mathbf{Z}|,$$

even if the law of $\mathbf{Z}$ is anisotropic.

With two centred independent Gaussian components instead,

$$\mathbb{E}[R^2] = a\tau^{2H_E} + b\tau^{2H_N},$$

unequal component exponents can produce a lag-dependent radial effective exponent.

East, north and radius together still do not identify the joint law

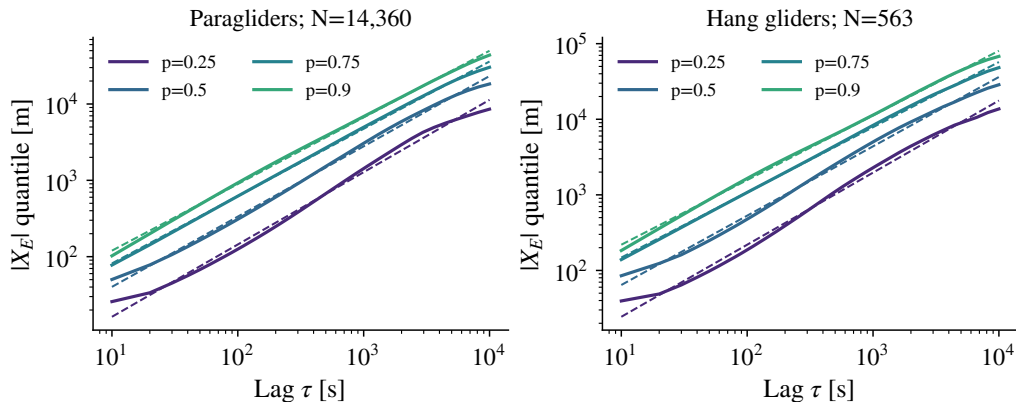
$$\{(1,2), (-1,-2)\} \quad \text{versus} \quad \{(1,-2), (-1,2)\},$$

each pair equiprobable.

Quantity	First law	Second law
Signed east marginal	± 1	± 1
Signed north marginal	± 2	± 2
Radius	$\sqrt{5}$	$\sqrt{5}$
Cross-covariance	$+2$	-2

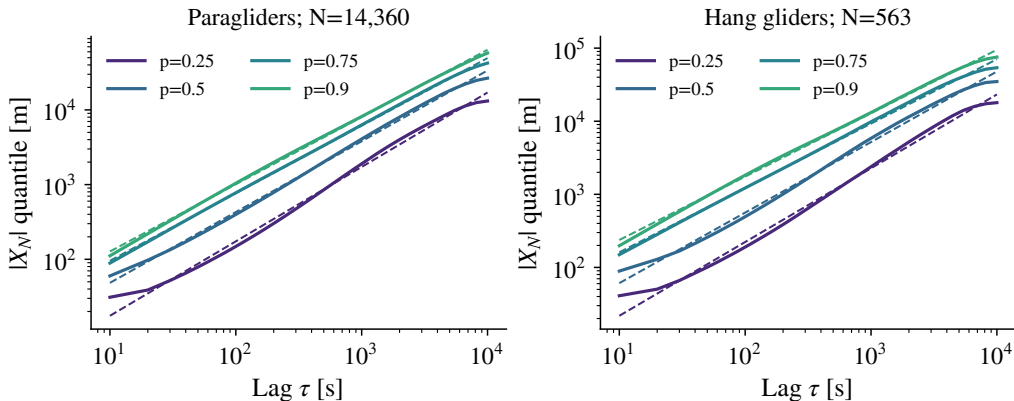
The missing information is real: signs and component dependence require a joint diagnostic.

Fixed-population quantiles: east component



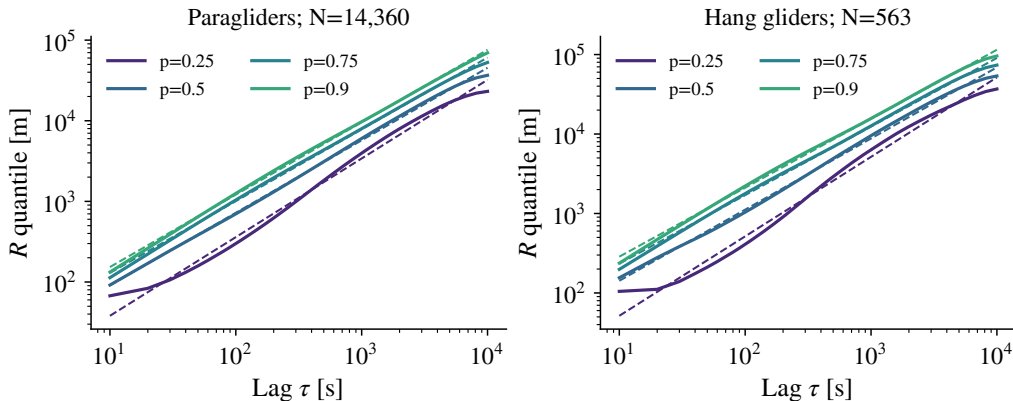
Solid: weighted empirical quantiles. Dashed: separate saved log-log fits over 10–10,000 s. Each dashed slope is one $H_{E,p}$.

Fixed-population quantiles: north component



Solid: weighted empirical quantiles. Dashed: separate saved log-log fits over 10–10,000 s. Each dashed slope is one $H_{N,p}$.

Fixed-population quantiles: radial displacement



Solid: weighted empirical quantiles. Dashed: separate saved log-log fits over 10–10,000 s. Each dashed slope is one $H_{R,p}$.

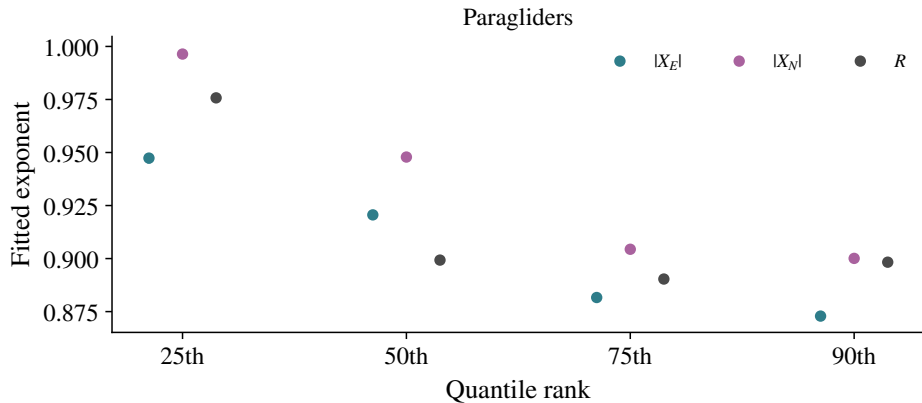
Whole-flight bootstrap preserves pairing across ranks, axes and lags

Resample flights with replacement 400 times. Each selected flight retains all its common origins and its joint east–north observations at every lag.

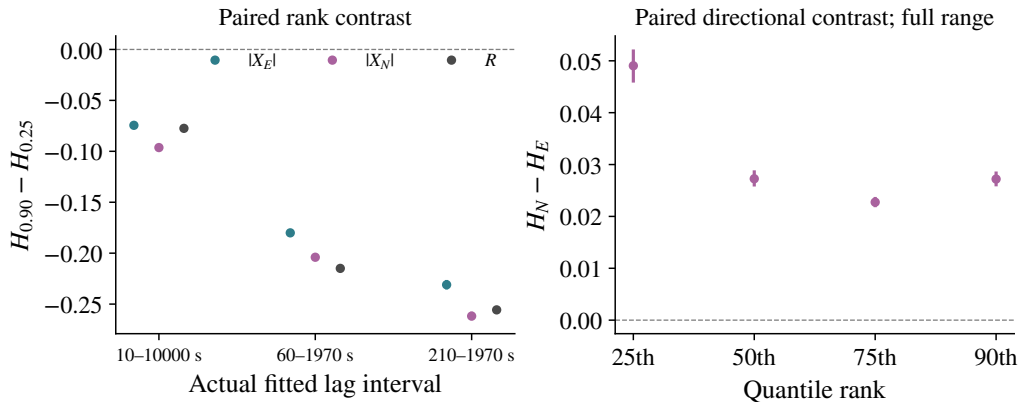
Recompute the weighted CDFs, extract all quantiles and repeat the same log–log fits in every draw. The 2.5th and 97.5th percentiles of the fitted slopes give the displayed 95% intervals; contrasts are computed within each draw.

Overlapping origins remain within the resampled flight; they are not treated as independent replicates.

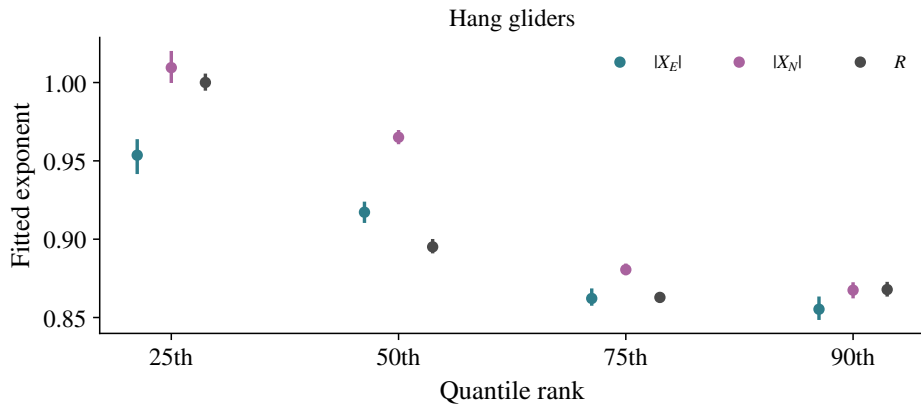
Paragliders: component and radial exponent estimates



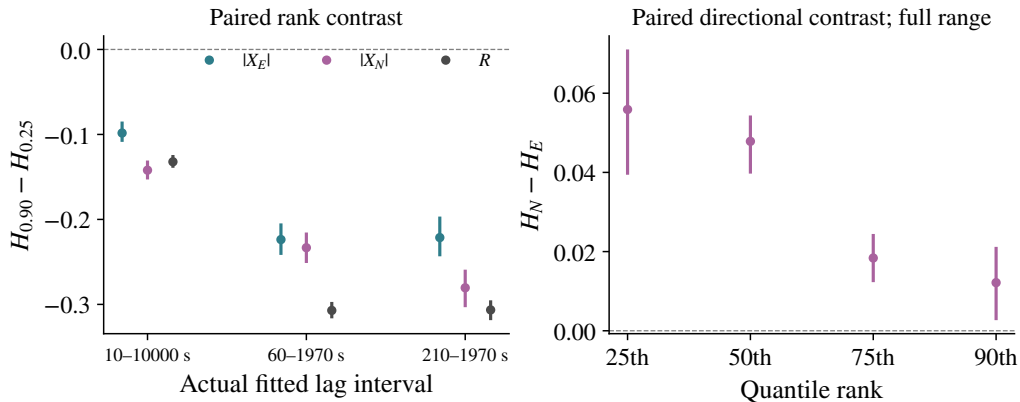
Paragliders: paired rank and directional contrasts



Hang gliders: component and radial exponent estimates



Hang gliders: paired rank and directional contrasts



Thesis §3.3.2

Distribution collapse

The collapse exponent shares one slope across four quantile ranks

The separate $H_{j,p}$ describe each curve. For the marginal collapse, fit one $H_{j,*}$ per quantity, with a separate intercept for each rank:

$$\min_{H, \{b_p\}} \sum_p \sum_{\ell \in \mathcal{L}} \left[\log \frac{Q_{j,p}(\tau_\ell)}{1 \text{ m}} - b_p - H \log \frac{\tau_\ell}{1000 \text{ s}} \right]^2.$$

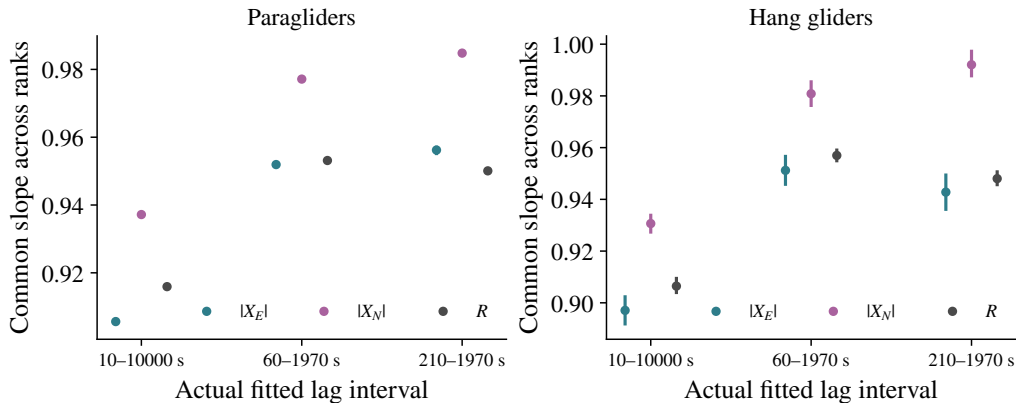
Equal rank and lag weights on the same grid give $H_{j,*} = \frac{1}{4} \sum_p H_{j,p}$. With $a_j(\tau) = [\tau / (1000 \text{ s})]^{H_{j,*}}$,

$$u = Y_j / a_j(\tau), \quad p_u(u; \tau) = a_j(\tau) p_{Y_j}(a_j(\tau)u; \tau).$$

The horizontal scale and vertical density change together to preserve probability. The exponent is fixed before comparing the rescaled laws.

Thesis Sec. 3.3.2. Separate $H_{j,p}$, shared marginal $H_{j,*}$ and the later joint-vector H_* serve different comparisons.

A common fitted scale depends on the fitted interval



A large density near zero need not carry much probability

$$\hat{f}_j = \frac{p_j}{b_{j+1} - b_j}.$$

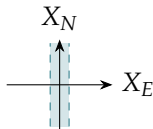
A very narrow histogram bin can have a tall density and still contain little probability. Across the displayed marginal laws, the mass below 1 mm, including zero and the straddling bin, is below 0.003%.

That numerical scale is not a claim of millimetre GNSS resolution. The following plots retain positive support; empirical-CDF comparisons also retain zero mass.

A radial density can scale without a plateau near zero

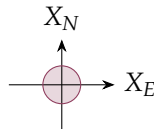
For a regular joint density, polar coordinates give

$$p_R(r; \tau) = r \int_0^{2\pi} p_X(r \cos \theta, r \sin \theta; \tau) d\theta.$$



A strip gives a component plateau:

$$p_{|X_E|}(x) \longrightarrow 2p_E(0) > 0.$$



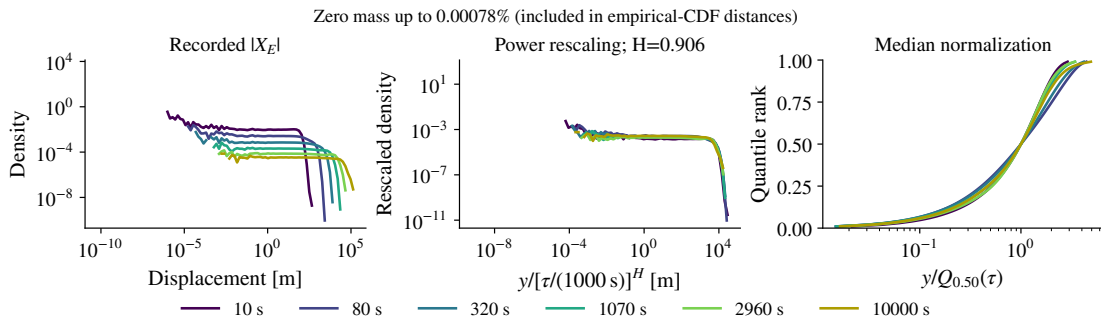
A disk gives a radial rise:

$$p_R(r) \sim 2\pi p_X(0,0)r.$$

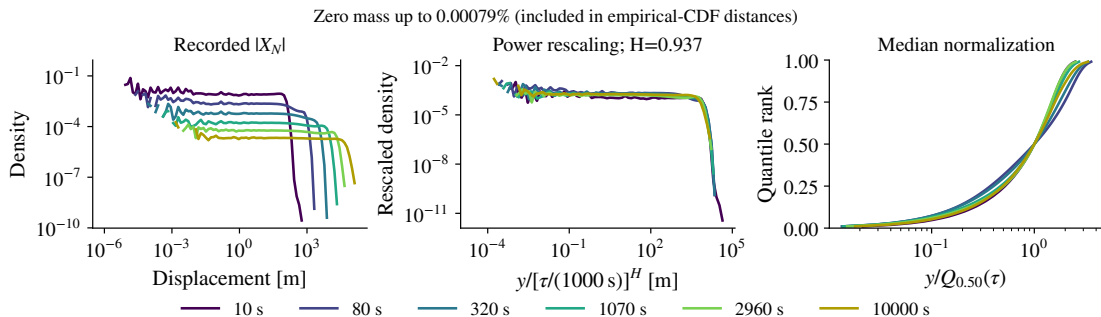
A rising radial density is compatible with exact scaling.

Finite, continuous, nonzero densities at the origin; no isotropy assumption. This explains the geometry, not a fitted small-radius law.

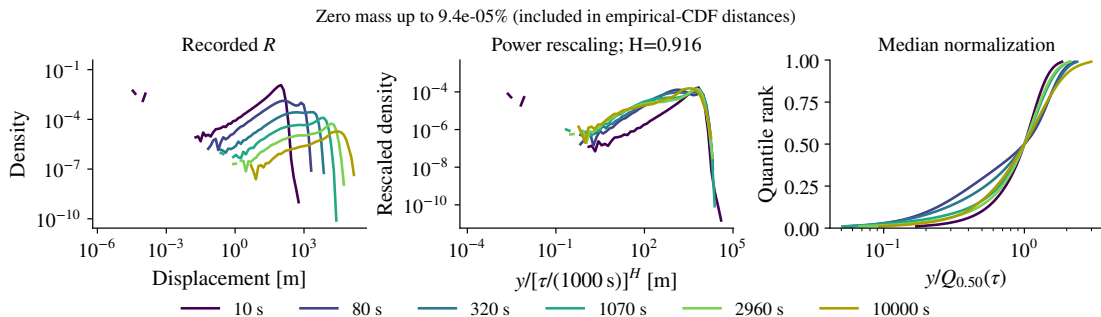
Paragliders: marginal collapse for east



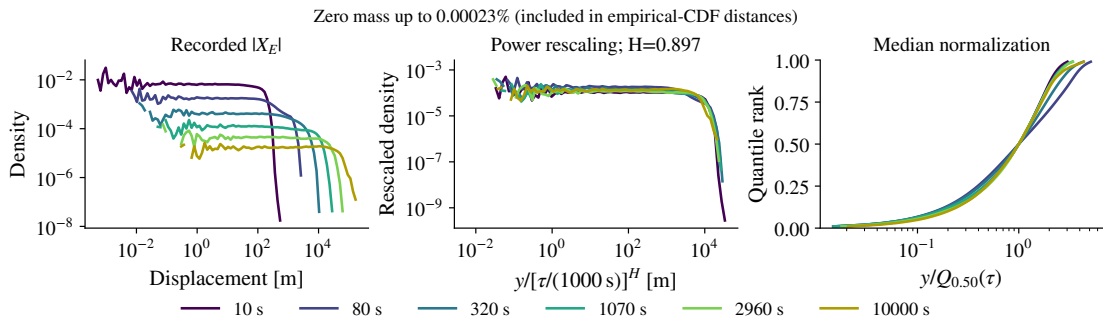
Paragliders: marginal collapse for north



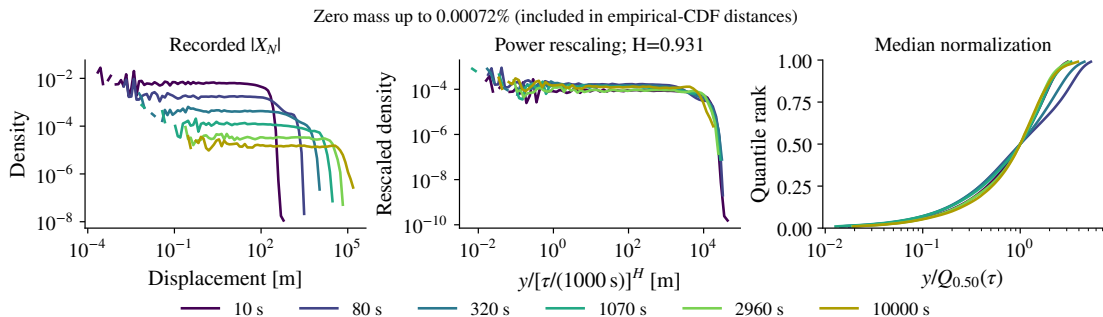
Paragliders: marginal collapse for radius



Hang gliders: marginal collapse for east

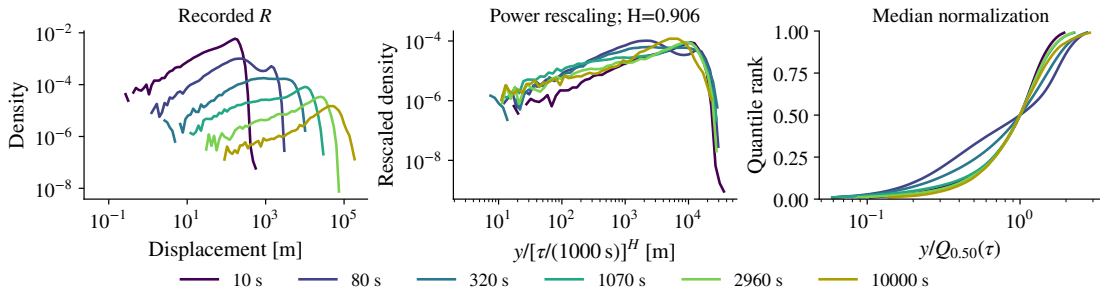


Hang gliders: marginal collapse for north



Hang gliders: marginal collapse for radius

Zero mass up to 0.00018% (included in empirical-CDF distances)



East and north show approximate collapse, with residual differences

Power rescaling visibly brings the absolute east and north densities together. Radial curves also group more closely, but differences remain around the peak and on its rising side.

Maximum ECDF separation	$ X_E $	$ X_N $	R
Paragliders	0.177	0.203	0.273
Hang gliders	0.185	0.269	0.331

A value of 0.177 means a 17.7-percentage-point cumulative probability difference for some pair of lags after power rescaling. Log density axes can conceal it.

Conclusion: approximate scaling of absolute-component one-lag laws. These plots do not establish signed-component or multi-time process self-similarity.

Thesis Figs. 3.10–3.11 and Table 3.5. Fixed flights, weights and origins; descriptive distances without a calibrated rejection threshold.

Thesis §3.3.3

Signed joint distributions

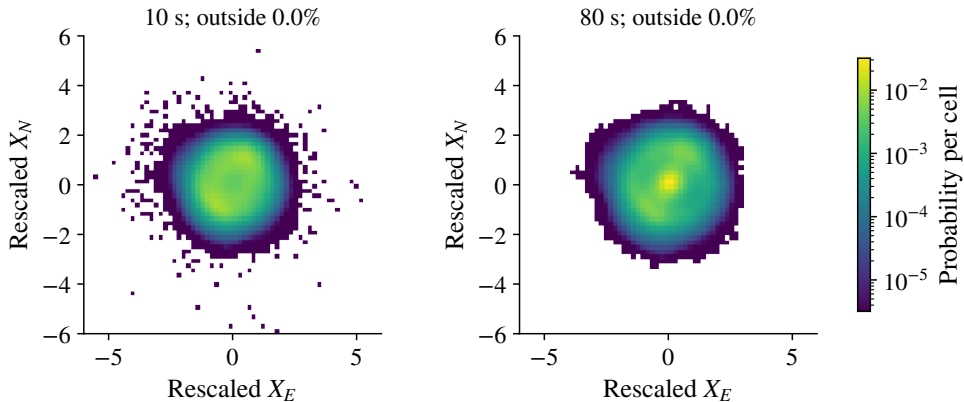
The joint comparison keeps geographic axes and one scalar scale

$$\mathbf{Z}_\tau = \frac{(X_E, X_N)}{s_0 \left(\frac{\tau}{1000\text{s}} \right)^{H_*}}.$$

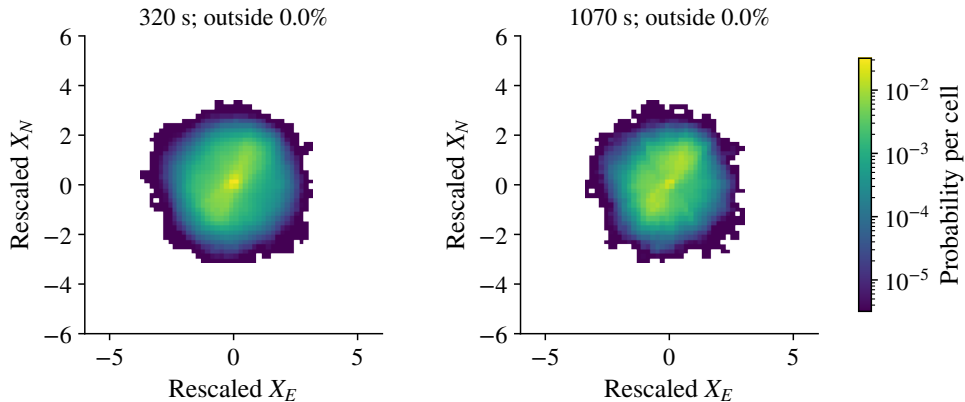
H_* is the common fit across all component/rank slopes. The radial reference median near 1000 s is adjusted to 1000 s by the same power.

There is no lag-specific centring, rotation or whitening. The six lags share the same cells and equal total weight per flight.

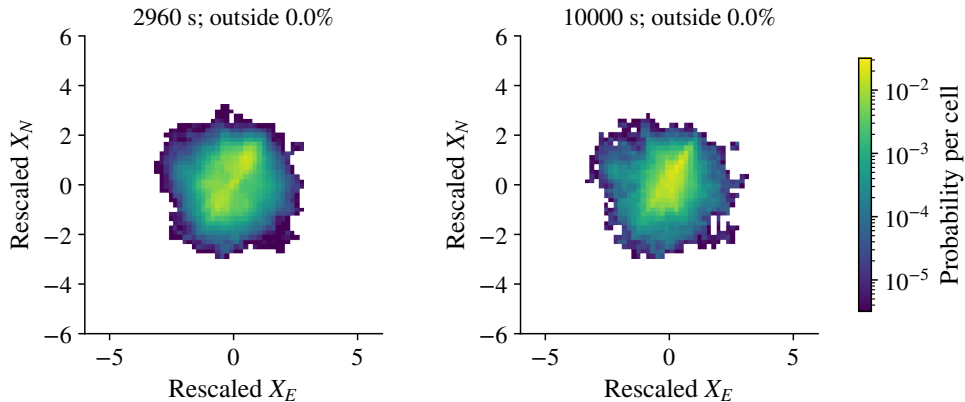
Paragliders: signed joint laws (1/3)



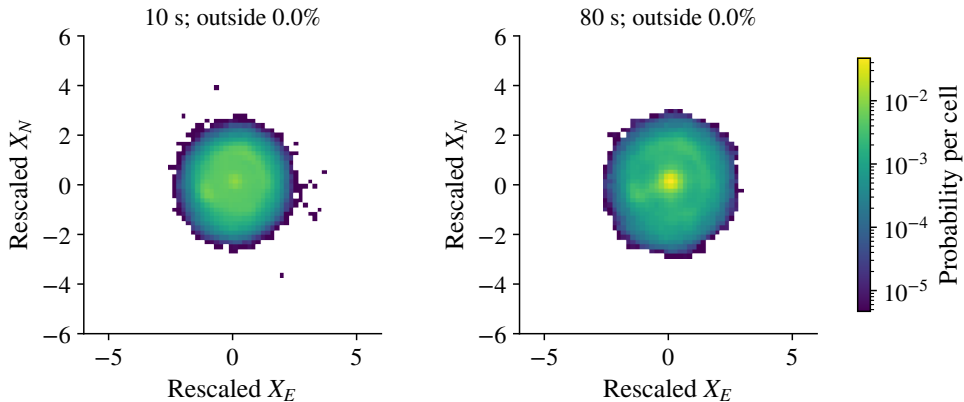
Paragliders: signed joint laws (2/3)



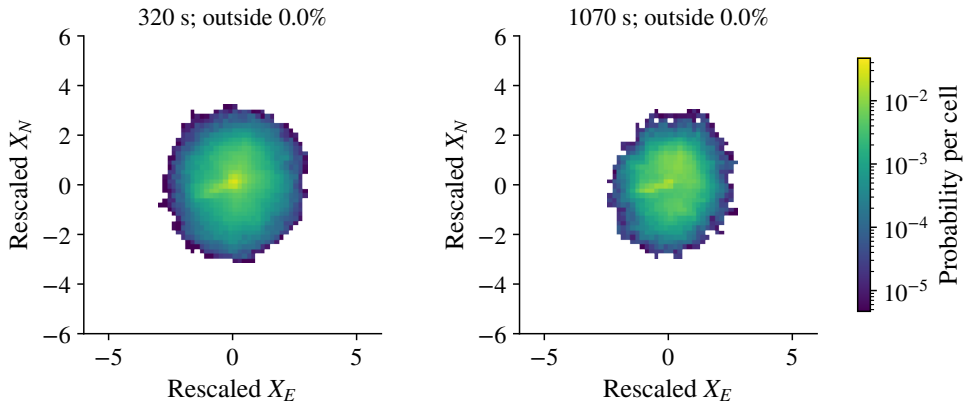
Paragliders: signed joint laws (3/3)



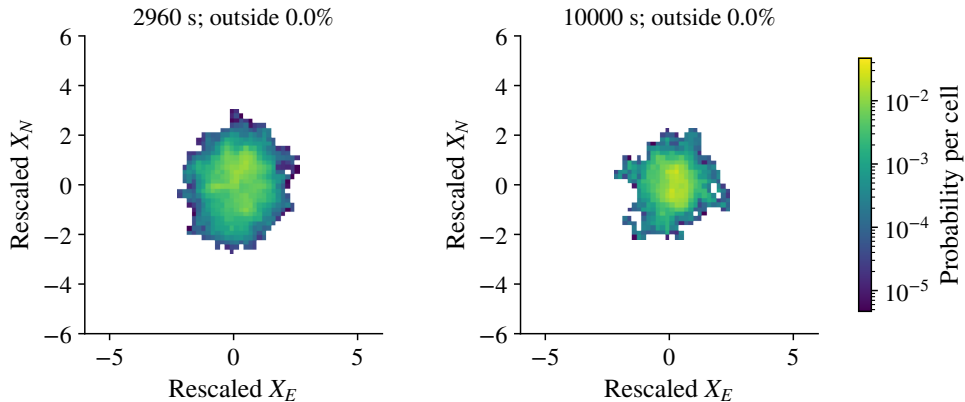
Hang gliders: signed joint laws (1/3)



Hang gliders: signed joint laws (2/3)



Hang gliders: signed joint laws (3/3)



One scalar rescaling leaves changes in the signed joint law

	Paragliders	Hang gliders
Common scalar exponent H_*	0.9196	0.9114
Binned TV: 10 s versus 10,000 s	0.370	0.403
Largest TV among the six lags	0.370	0.441

Later rescaled laws concentrate more centrally, while their means also change. Overflow mass is zero to numerical precision in the displayed square.

Mean motion, centred spread, orientation and dependence can all contribute. The plot alone does not separate their physical causes.

$$D_{\text{joint}}(\tau, \tau') = \frac{1}{2} \sum_{ij} |p_{\tau,ij} - p_{\tau',ij}|.$$

The grid has 64 interior bins per axis and overflow cells. The distance resolves changes at that bin scale; it is not a calibrated rejection test.

Even agreement of every one-lag vector distribution would leave the joint law at several times, and hence full process self-similarity, untested.

Thesis §3.3.4

Testing approximate scaling beyond one-lag magnitudes

Two consecutive displacements test temporal scaling

$$\mathbf{X}_1 = \mathbf{r}(t + \tau) - \mathbf{r}(t), \quad \mathbf{X}_2 = \mathbf{r}(t + 2\tau) - \mathbf{r}(t + \tau),$$

$$\mathbf{Z}_\tau = (X_{1,E}, X_{1,N}, X_{2,E}, X_{2,N}), \quad \tilde{\mathbf{Z}}_\tau = \frac{\mathbf{Z}_\tau}{s_0(\tau/\tau_0)^{H_*}}.$$

An H -self-similar process with stationary increments predicts the same rescaled law at every lag.

$(\mathbf{X}_1, \mathbf{X}_2)$ also determines positions relative to the origin at τ and 2τ : $(\mathbf{X}_1, \mathbf{X}_1 + \mathbf{X}_2)$.

Signs and the association between consecutive intervals are retained.

Keep common support and separate training from validation

Lag grid (s)	Support through	Eligible para / hang
10, 100, 1000, 10000	20000 s	14360 / 563
100, 200, 400, 800, 1600	3200 s	149831 / 5914
1000, 2000, 4000, 8000	16000 s	26977 / 1240

Every supported 10-s origin in every eligible segment; identical origins at all lags within a grid. Equal total weight per flight.

A fixed calendar-date split assigns whole days to training or validation. On training days, fit the four quantile ranks of $|E_1|$, $|N_1|$ and R_1 ; average their twelve log-log slopes to obtain one H_* . Use the training radial median at the smallest lag as s_0 .

The exponent and length scale are then fixed on validation days. No per-lag centring, rotation or whitening.

Measure probability discrepancies and their uncertainty

Four coordinate axes and all twelve normalised pairwise sums and differences give 16 fixed signed projections u . Evaluate their CDFs at

$$z \in \{0, \pm 0.1, \pm 0.25, \pm 0.5, \pm 1, \pm 2, \pm 4, \pm 8\}.$$

$$\hat{D} = \max_{\tau < \tau', u, z} |\hat{F}_{\tau, u}(z) - \hat{F}_{\tau', u}(z)|.$$

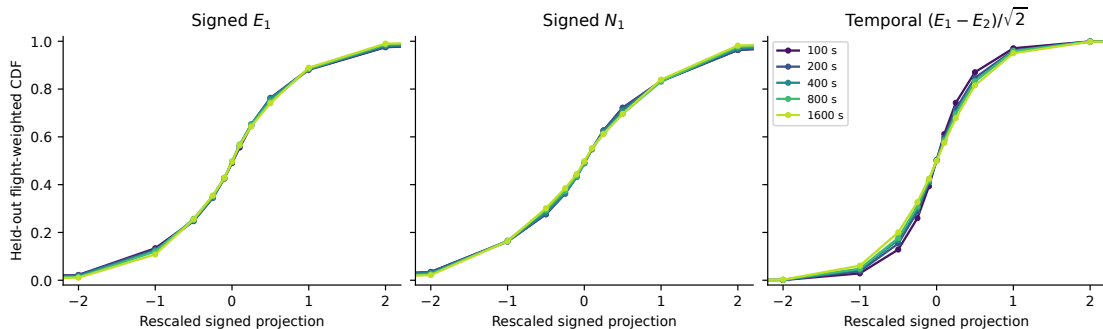
Resample whole validation dates 999 times, keeping the same multiplicity for every flight, origin, lag and projection within a day.

The 95th percentile $c_{.95}$ of the maximum centred bootstrap error gives the upper bound $U = \min(1, \hat{D} + c_{.95})$.

Require U below a declared tolerance: 0.05, 0.10 or 0.20.

Simultaneous within one discipline and grid; conditional on the training fit. Finite projections and thresholds, not an omnibus 4D test.

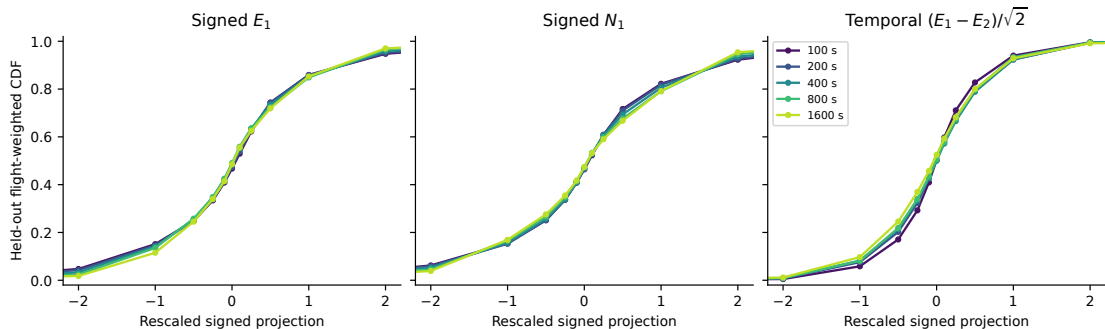
Paragliders: intermediate scaling survives a temporal check



100–1600 s; training $H_* = 0.915$. Spatial-projection upper bound: 0.0313. Temporal upper bound: 0.0829.

The declared 0.05 spatial and 0.10 temporal tolerances are supported.

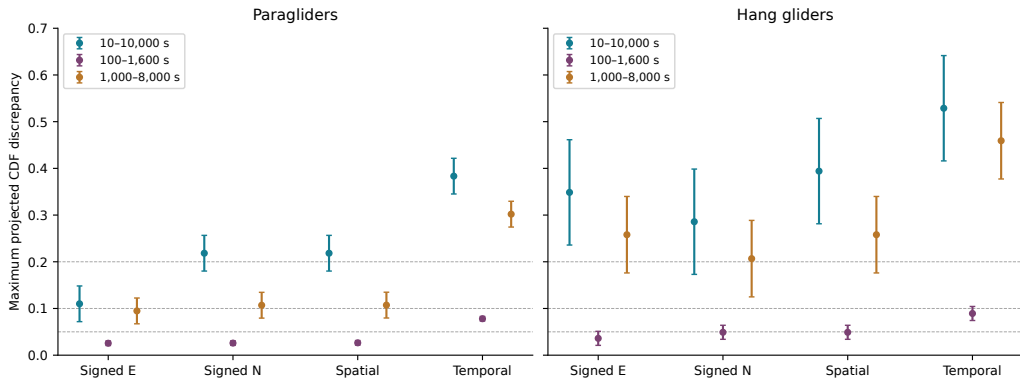
Hang gliders: similar collapse, less precise temporal agreement



100–1600 s; training $H_* = 0.899$. Spatial upper bound: 0.0640. Temporal distance: 0.0893; upper bound: 0.1043.

Spatial agreement passes 0.10; the temporal result narrowly misses it.

The scaling approximation has a restricted lag range



On the full and long grids, temporal lower bounds exceed 0.20 in both disciplines.

Points: maximum CDF differences. Bars: simultaneous 95% bootstrap bounds. Dashed lines: declared tolerances. Each grid has its own fixed cohort.

The supported claim is approximate intermediate-range scaling

Evidence strengthened beyond visual collapse

Over 100–1600 s, one scalar exponent approximately preserves signed spatial and tested two-interval laws, most clearly for paragliders.

This supports an approximate self-similar description at the measured directions, thresholds, tolerances and population.

Exact invariance, arbitrary multi-time laws and increment stationarity remain unestablished. The fitted exponent is not an intrinsic Hurst exponent.

An intermediate interval between short manoeuvres and longer route constraints is a plausible interpretation; the mechanisms are not identified here.

Thesis §3.4.1

Flight weighting and a fixed-segment control

Compare transport magnitude before its directional structure

Two groupings

Alps, Pyrenees and Channel Coast use the defined geographic boxes.

Initial GNSS altitude h_0 defines:

Plains	$h_0 < 300$ m
Hills	$300 \leq h_0 < 800$ m
Low mountains	$800 \leq h_0 < 1500$ m
High mountains	$h_0 \geq 1500$ m

One flight-level estimator

$$m_f(\tau) = \frac{\sum_s n_{fs} m_{fs}}{\sum_s n_{fs}},$$

$$M_g(\tau) = \frac{1}{N_g} \sum_{f \in g} m_f(\tau).$$

Pool supported origins inside each flight, then weight contributing flights equally.

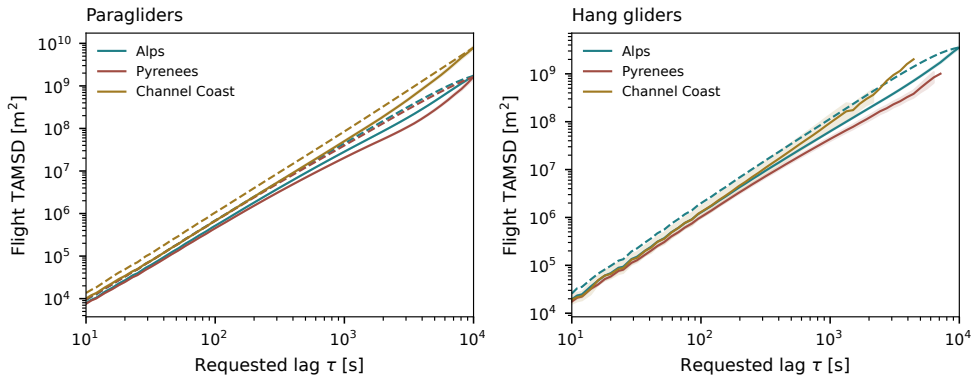
Compare available segments with a fixed set supporting the largest lag.

Native-grid TA-MSD; 155,085 para / 6,060 hang flights. Altitude groups include the full eligible archive.

Thesis §3.4.2

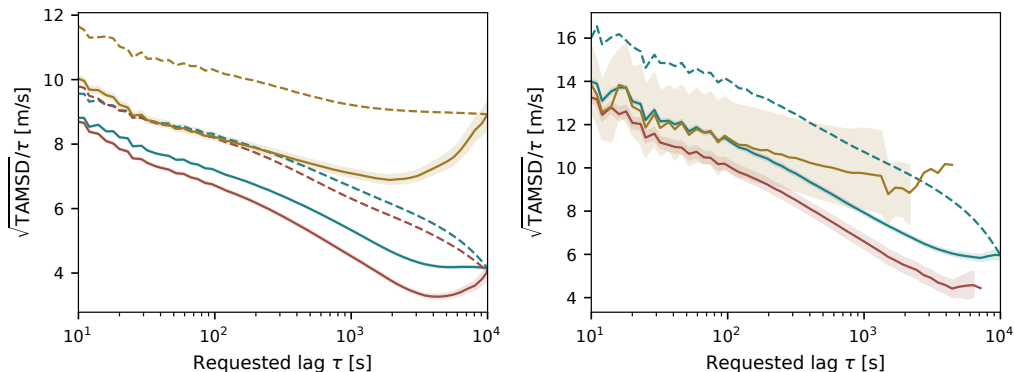
Regional contrasts and changing population

Regional TA-MSD: coastal flights travel farther at intermediate lags



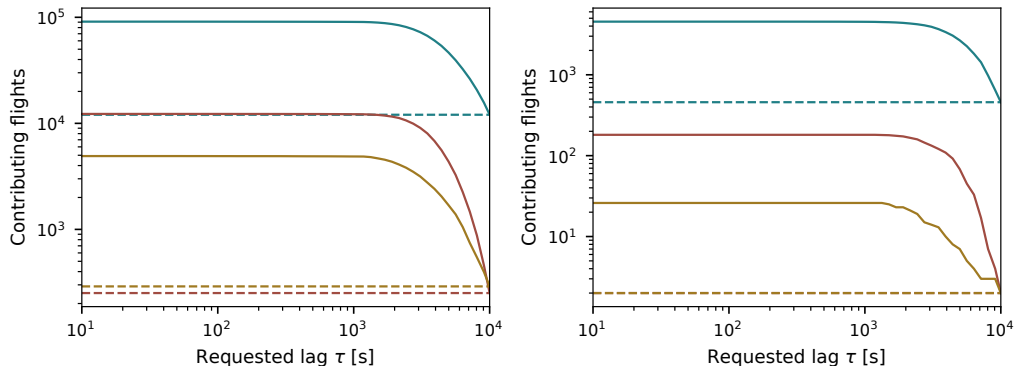
Solid: available segments. Dashed: fixed long segments. Shading: pointwise 95% site-day bootstrap intervals for solid curves.

Fixing the population changes the apparent long-lag growth



The ordinate is RMS interval-mean ground speed, up to native-grid rounding. A horizontal curve corresponds to quadratic TAMSD growth.

Regional support limits what the fixed control can establish

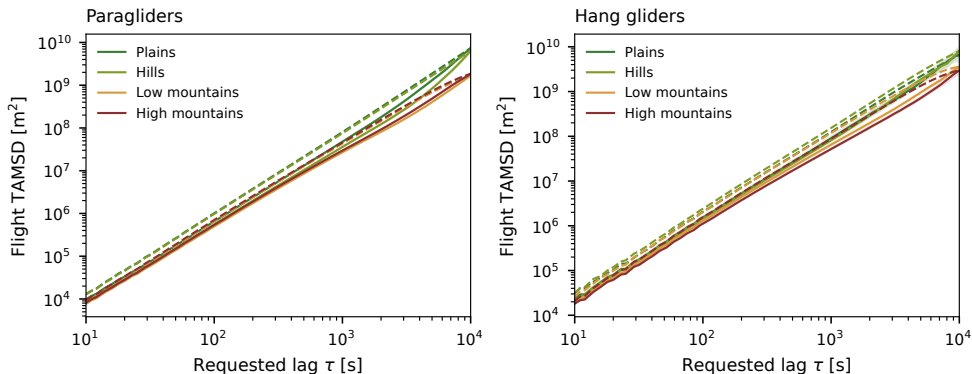


Fixed paraglider flights: 12,053, 251, 290.

Thesis §3.4.3

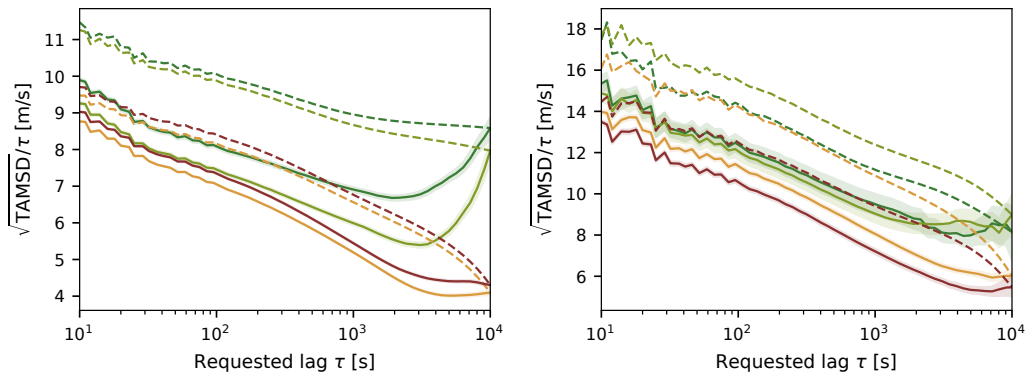
Initial-altitude bands and their regional composition

Initial-altitude groups combine transport and regional composition



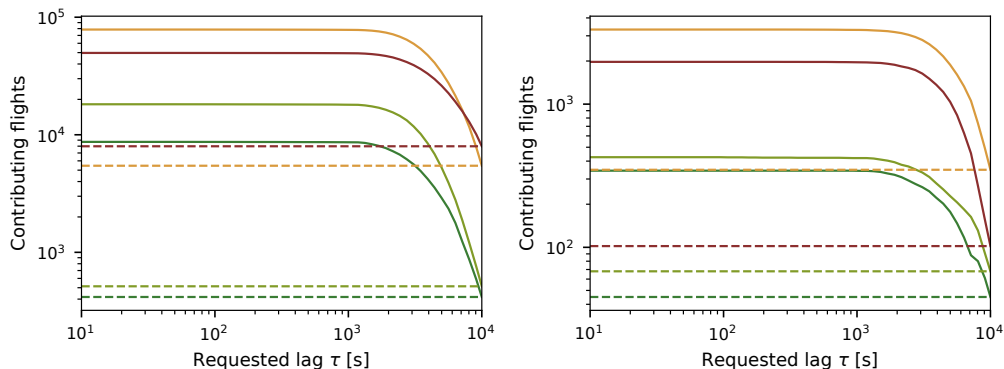
Bands use the cleaned GNSS origin: Plains, Hills, Low mountains, High mountains. All eligible regions enter these curves.

The late rise in Plains and Hills is sensitive to population selection



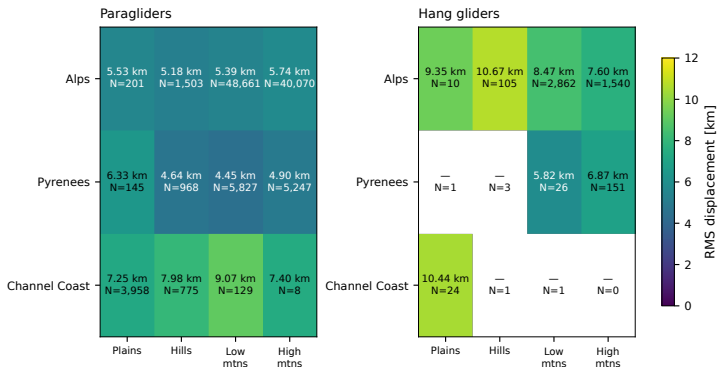
Solid: available segments; dashed: identical long segments at every lag. Shading is pointwise, not simultaneous.

The altitude-band comparison also needs its flight counts



Counts concern supported flights. The fixed-segment control retains membership; its origins still depend on lag.

Region and origin altitude are strongly associated in this sample



RMS displacement at the requested 1049-s lag; available segments. Descriptive cells, with no calibrated intervals.

Thesis §3.5

Directional structure, terrain and equipment

Regional plots use different clocks, weights and centring

Diagnostic	Clock	Weighting	Centring
Launch-time ratio	time since launch t	paired flights at each t	raw moments
Regional PCA	increment lag τ	eligible origins pooled	mean removed
Fixed joint law	increment lag τ	equal total flight weight	mean retained

These measurements differ in method as well as population. Compare these choices before attributing a difference to a physical mechanism.

Thesis §3.5.1

East–north balance in the retained archive

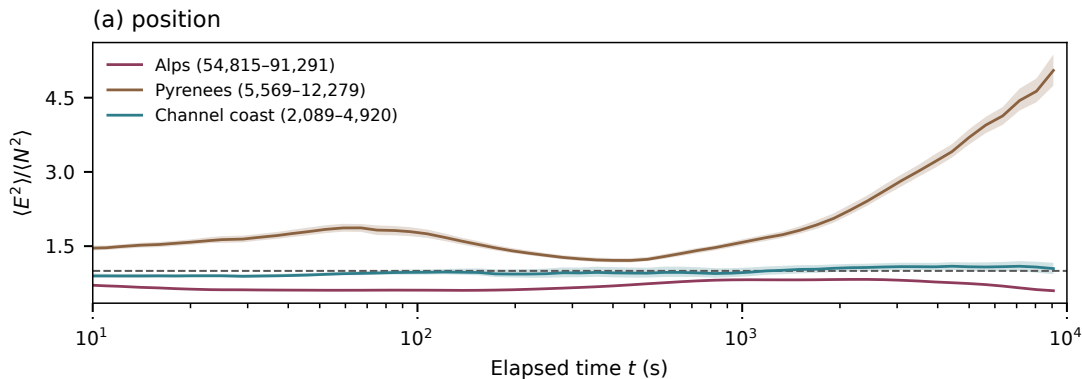
Equal east and north moments do not establish isotropy

$$\rho_{EN} = 1, \quad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \lambda_1/\lambda_2 = 19.$$

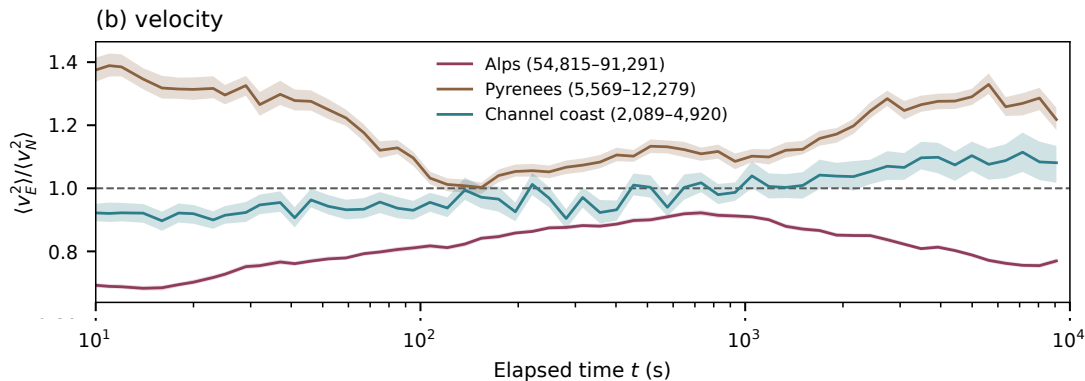
This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

Regional position moments differ along east and north



Regional velocity moments show a related directional contrast



The ratios use paired east and north observations. A site–day bootstrap with 200 draws gives pointwise 10–90% ranges; each plotted time needs at least 30 flights and 10 clusters.

Missing site/date keys are treated as singletons. Resampling these groups does not match the equipment populations on weather, task or duration.

Thesis §3.5.2

Regional principal axes

PCA describes preferred axes; wind needs independent information

A rotation preserves distance

$$\Sigma = U\Lambda U^\top, \quad |U^\top \mathbf{X}| = |\mathbf{X}|.$$

PCA decorrelates components at a chosen lag. It cannot repair radial collapse or remove temporal dependence.

Whitening changes lengths and only fixes second moments at the fitted scale.

The following panels report regional PCA and its support. Anisotropy alone is compatible with a self-similar vector and a self-similar radius.

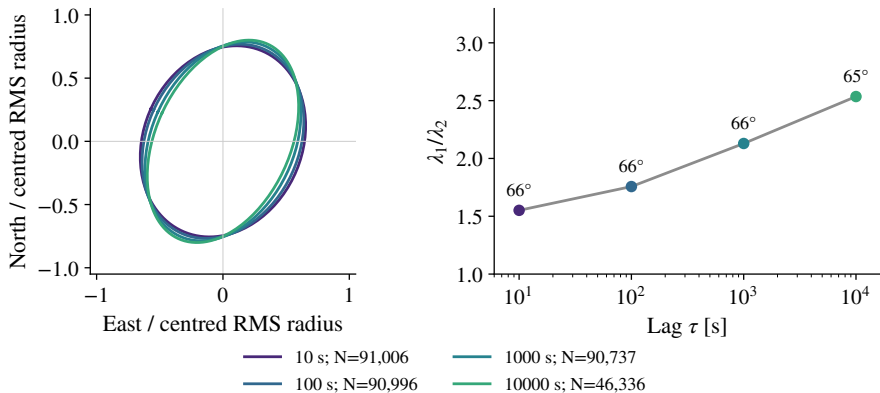
Ground velocity has two unknown terms

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

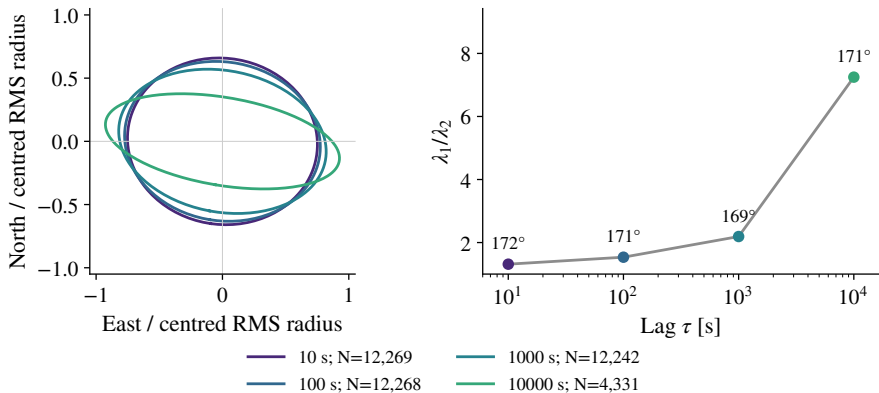
Terrain constrains headings; wind also changes position. Differentiation does not separate the two.

Different marginal exponents, or changing joint dependence, can prevent radial scaling.

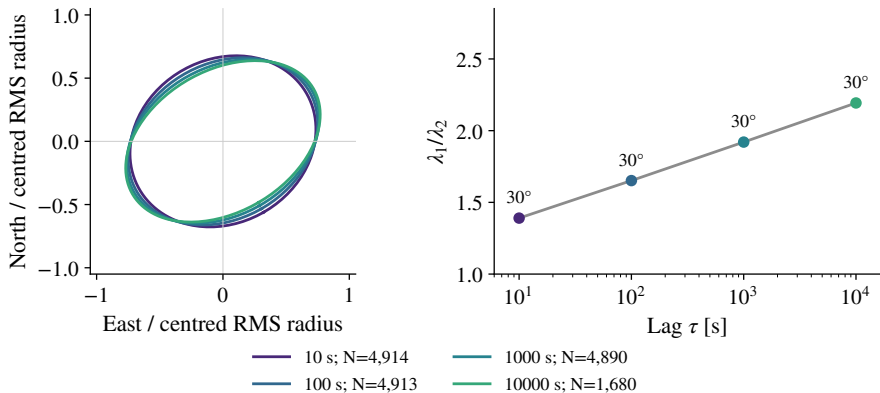
Alps: centred principal axes across four lags



Pyrenees: centred principal axes across four lags



Channel Coast: centred principal axes across four lags



Directional constraints become more visible at longer lags

λ_1 / λ_2	10 s	100 s	1000 s	10,000 s
Alps	1.55	1.76	2.13	2.54
Pyrenees	1.31	1.54	2.19	7.25
Channel Coast	1.39	1.65	1.92	2.19

The centred displacement cloud becomes more elongated over longer intervals. Both principal variances may still grow.

Physical interpretation: longer increments can reveal route constraints and persistent wind contributions that are less visible over a short part of a flight.

Here τ is increment duration. The four measured lags do not establish a positive derivative at every scale; the contributing flights also change.

Thesis Section 3.5.2 and Fig. 3.21. Regional paraglider PCA; the environmental mechanisms remain hypotheses.

Terrain constraints appear as a route explores their scale

τ_{terrain} is an illustrative time to explore a distance L across a valley or flight corridor; it has not been measured here.

When an increment spans such a distance, terrain or the edge of usable lift may require a turn. Across-corridor displacement is limited, while motion along the corridor can continue.

$$\tau_{\text{terrain}} \sim \frac{L}{v_{\perp}} \quad (\text{persistent crossing}), \quad \tau_{\text{terrain}} \sim \frac{L^2}{D_{\perp}} \quad (\text{diffusive exploration}).$$

$v_{\perp}$: transverse speed; $D_{\perp}$: transverse diffusion coefficient.

The two estimates assume different types of motion. Pilots may anticipate terrain, and local lift can affect headings at shorter lags.

Thesis Section 3.5.2, environmental interpretation. Actual terrain constraints depend on altitude, lift and route choice.

A weak persistent wind contribution can emerge with lag

Wind displacement accumulates as $\Delta \mathbf{r}_w(t, \tau) = \int_0^\tau \mathbf{w}_h(t+s) ds$. Direction changes can cause cancellation; persistence allows coherent accumulation.

Illustration: an isotropic background with covariance $2D\tau I_2$, plus an independent wind along $\mathbf{e}$, varying across observations with variance σ_w^2 and constant within each increment:

$$\Sigma(\tau) = 2D\tau I_2 + \sigma_w^2 \tau^2 \mathbf{e}\mathbf{e}^\top, \quad \frac{\lambda_1}{\lambda_2} = 1 + \frac{\sigma_w^2 \tau}{2D}.$$

A weaker contribution needs a longer interval to compete with background spread, provided it stays coherent that long.

A constant wind shared by all observations shifts only the mean and cancels from this centred PCA.

Thesis Section 3.5.2, illustrative wind model (not fitted). Temporal-correlation relation: Taylor (1922).

Whitening each lag makes each increment covariance isotropic

$$W_\tau = \Sigma_g(\tau)^{-1/2}, \quad \mathbf{Z}_\tau = W_\tau[\mathbf{X}_\tau - \boldsymbol{\mu}_g(\tau)].$$

$$\text{Cov}(\mathbf{Z}_\tau) = W_\tau \Sigma_g(\tau) W_\tau^\top = I_2.$$

At each lag, every unit-vector projection has variance one. The covariance ellipse is a circle by construction.

This requires a positive-definite covariance. With estimated matrices, the identity holds on the same weighted observations used for fitting.

Angular structure beyond covariance and dependence between times remain to be tested.

Thesis Section 3.5.2. Whitening identity: Kessy, Lewin and Strimmer (2018), doi:10.1080/00031305.2016.1277159.

A terrain direction measured independently of the flights

NOAA ETOPO 2022

Same Alpine and Pyrenean take-off boxes.

One-arc-minute elevation sampling; retain the footprint above 1000 m.

Project cell centres to regional east/north coordinates. The area-weighted spatial covariance gives the main axis.

[NOAA NCEI, ETOPO 2022](#). Both elevation subsets, requests, code and hashes are archived. Angles are counterclockwise from east, modulo 180° .

Definition sensitivity

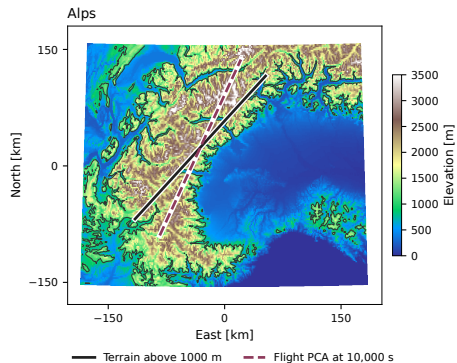
Thresholds: 500, 1000, 1500 m.

Alps: $42.1\text{--}47.9^\circ$.

Pyrenees: $169.7\text{--}175.4^\circ$.

These spans are not confidence intervals; the axis describes an area, not a traced crest.

Alps: shared sector, with an approximately 18-degree offset



Terrain: 47.6° .

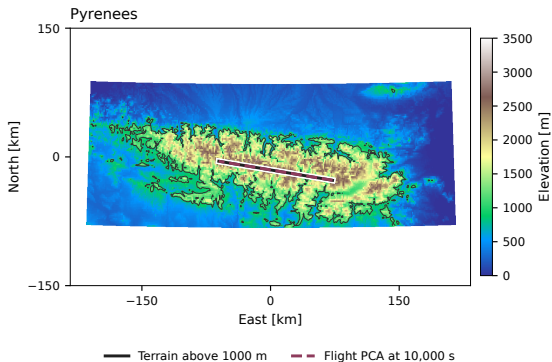
Flight PCA: 65.2° .

Separation: 17.6° .

46,336 flights support 10,000 s.

Black: highland-footprint axis. Dashed colour: current flight PCA, shifted to the same centre. Line lengths have no physical meaning.

Pyrenees: the long-lag spread follows the highland axis



Terrain: 170.5° .

Flight PCA: 170.5° .

Separation: less than 1° .

4,331 flights support 10,000 s.

ETOPO footprint above 1000 m; threshold sensitivity: $169.7\text{--}175.4^\circ$. Decimal agreement does not imply this physical precision.

Channel Coast: the reference height follows the PCA windows

Same population as the 10,000-s PCA

- 1,680 paraglider flights.
- 2,013 nonoverlapping windows, wholly inside retained segments.
- Mean cleaned GNSS altitude along each window; equal window weights.

Mean: 1,006 m

Equal flight weights: 980 m.

Middle half of window means:
854 m–1,160 m.

Approximate altitude above sea level;
recorder datums are not harmonised.

Trapezoidal integration on the same 10-s segment grids. The target describes the supported long-lag windows; it is not take-off altitude or height above the terrain.

ERA5 wind is interpolated to the measured flight altitude

2016–2025, hourly; nine cells in the coastal box; 1000/925/850/700 hPa.

$$H = \Phi / g_0, \quad h = \frac{R_{\oplus} H}{R_{\oplus} - H}, \quad \theta_w = \text{atan2}(\langle v(h) \rangle, \langle u(h) \rangle).$$

Convert geopotential to each hour's level heights. Interpolate **E/N wind components** to 1,006 m, between above-ground bounds.

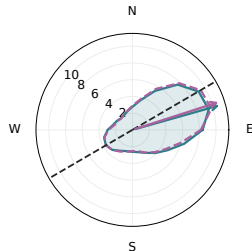
Main comparison: all year, all hours vs. all year, 09–17 UTC.

April–September is a separate seasonal check. Winds at 10/100 m AGL provide lower-level controls.

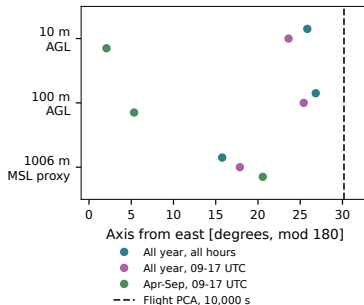
[ECMWF/Copernicus ERA5 pressure levels](#), accessed through a pinned snapshot of [Earthmover's public cloud-optimised ERA5 copy on AWS](#). Earthmover distributes this lossless copy; it is not a separate reanalysis. Complete hourly extract: 32.22 MB compressed, preserved in the repository and on SSD.

At flight height, both annual references remain offset from PCA

At mean flight altitude: 1006 m
Towards-sector frequency [%]



Height and time selection



Annual gaps at flight height: 14.4° for all hours, 12.3° for 09–17 UTC.

Angles counterclockwise from east. The rose shows where air moves towards at 1,006 m; the black line is the 30.2° centred PCA axis.

Height and time checks preserve a broad orientation resemblance

At 1,006 m	Wind axis	PCA gap	ρ_w
All year, all hours	15.8°	14.4°	0.43
All year, 09–17 UTC	17.9°	12.3°	0.44
Apr–Sep, all hours	16.0°	14.2°	0.35
Apr–Sep, 09–17 UTC	20.6°	9.6°	0.37

At the quartile heights, annual axes span 14.3–17.2° (all hours) and 16.4–19.3° (daytime), on shared support.

$\rho_w = \|\langle \mathbf{w} \rangle\| / \langle \|\mathbf{w}\| \rangle$ measures cancellation. Height ranges are sensitivities, not confidence intervals. The main estimates retain all selected hours.

What the environmental comparisons establish

Region	Conclusion supported by this comparison
Pyrenees	Close terrain–flight alignment; a geographical association worth testing along tracks.
Alps	Same broad sector, with an approximately 18-degree offset from the mean highland axis.
Channel Coast	Wind at mean flight altitude lies about 12–14 degrees from PCA in both annual time selections; seasonal checks retain a broad resemblance.

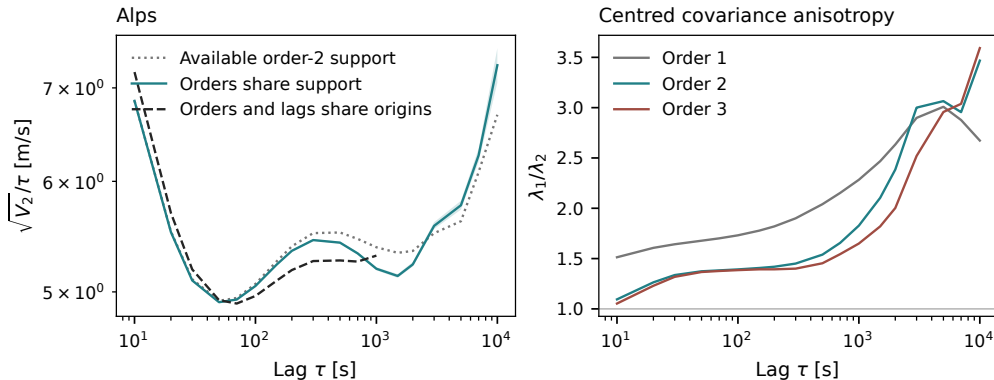
A common constant wind shifts the mean displacement by $\tau\mathbf{w}$; centring removes it. Variable winds and route responses can still shape covariance.

Independent regional references constrain an interpretation. They do not separate wind, terrain, route choice and long-flight selection.

Thesis §3.5.2

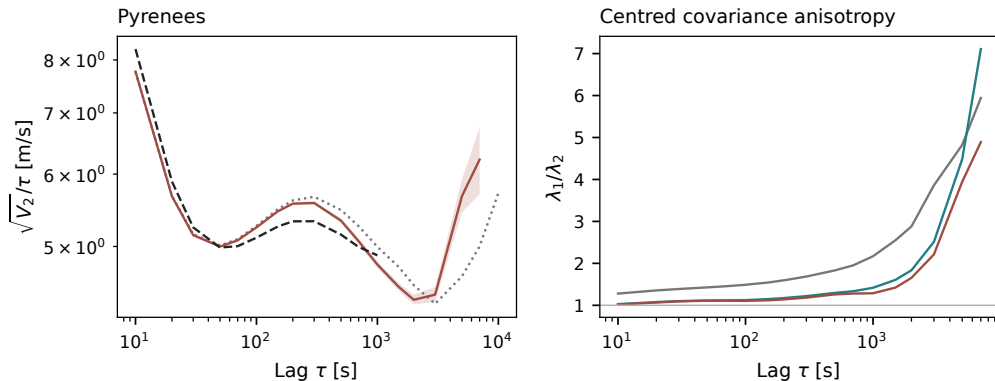
Regional changes of velocity and direction

Alps: velocity changes retain a directional structure



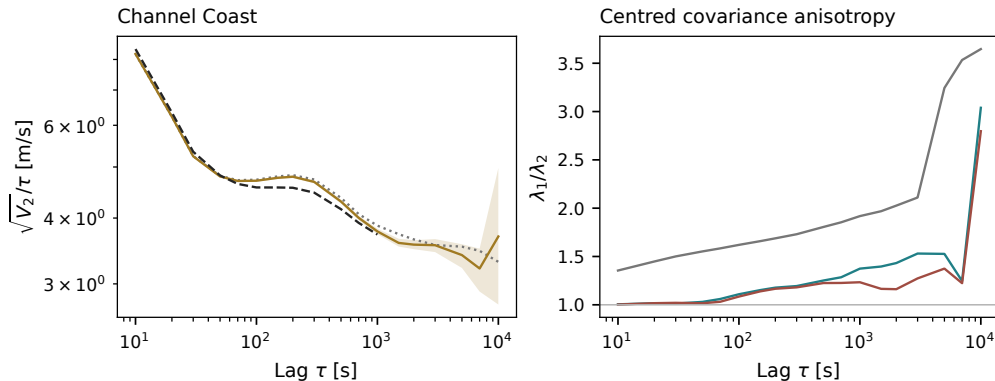
Equal flight weights. At 1000 s, matched orders retain 89,181 flights. Shading: pointwise 95% site-day intervals.

Pyrenees: the long-lag change coincides with falling support



At 1000 s: 12,020 matched flights; at 10,000 s: only 3. Cells below eight flights are omitted from plots.

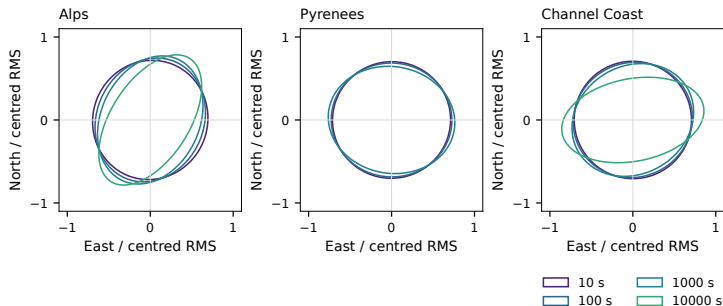
Channel Coast: smaller changes at 1000 s, with residual anisotropy



At 1000 s: 4,679 matched flights, $D_2 = 3.77$ m/s and covariance ratio 1.37.

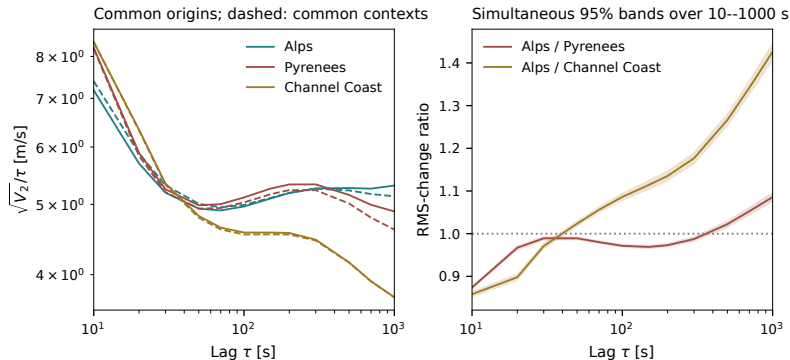
Second-difference ellipses retain orientation after drift cancellation

$$C_2 = \text{Cov}(A_2 \mathbf{r}), \quad V_2 = \text{tr } C_2 + |\langle A_2 \mathbf{r} \rangle|^2.$$



Matched-order origins, equal flight weights; each ellipse is normalised by its own centred RMS.

The regional contrast persists in a common-context control

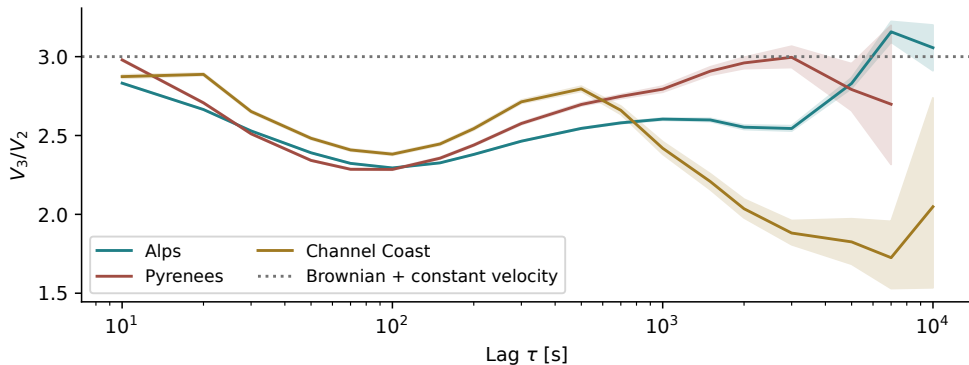


Common origins through 1000 s. Dashed: 23 shared context strata, descriptive. Shading: simultaneous 95% bands for the unstandardised ratios.

Thesis §3.5.2

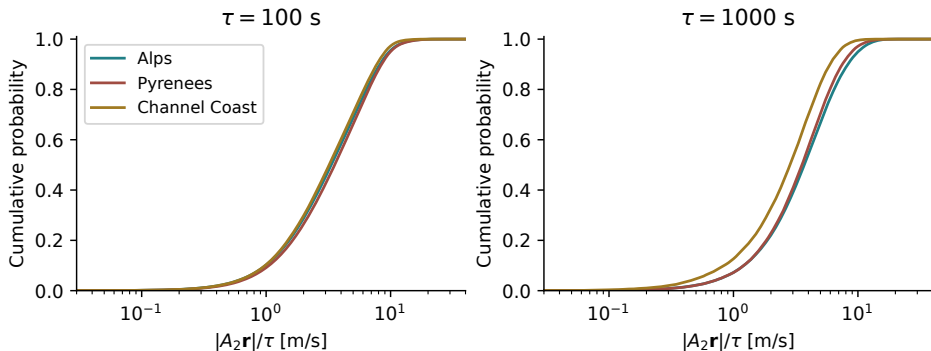
Third differences and the distribution of changes

The ratio between orders changes with the lag



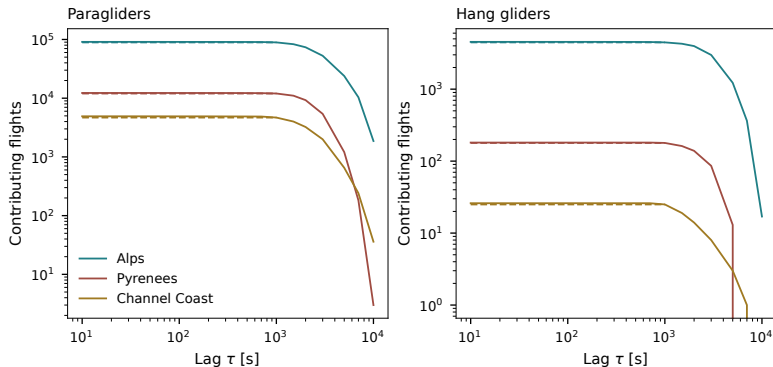
Matched origins and flight weights; pointwise 95% site-day intervals. Dotted: Brownian motion plus constant velocity.

The distribution distinguishes frequent small changes from large events



Equal-flight CDFs on matched-order support. Full report: all four display lags, zero and overflow mass, and quantile bin bounds.

Order-three support limits the long-lag regional comparison

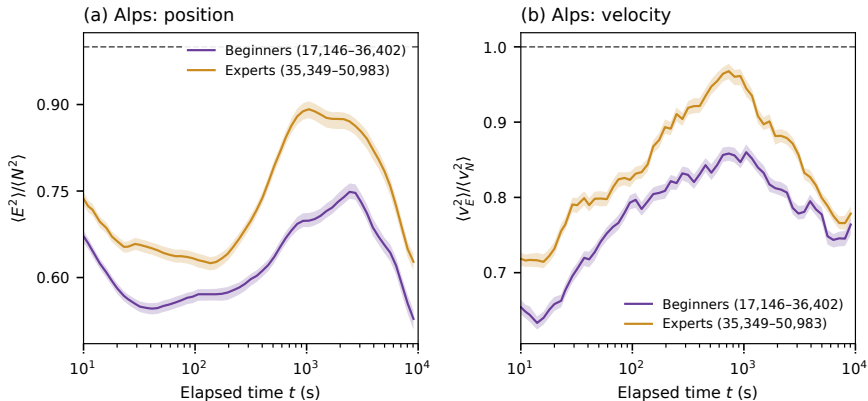


Solid: orders share 3τ support. Dashed: origins common through 1000 s. All eligible regional flights contribute where supported.

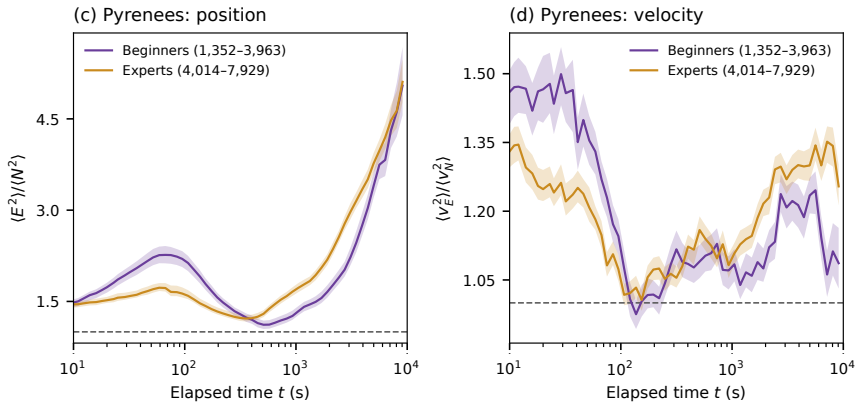
Thesis §3.5.3

Terrain, equipment and wind

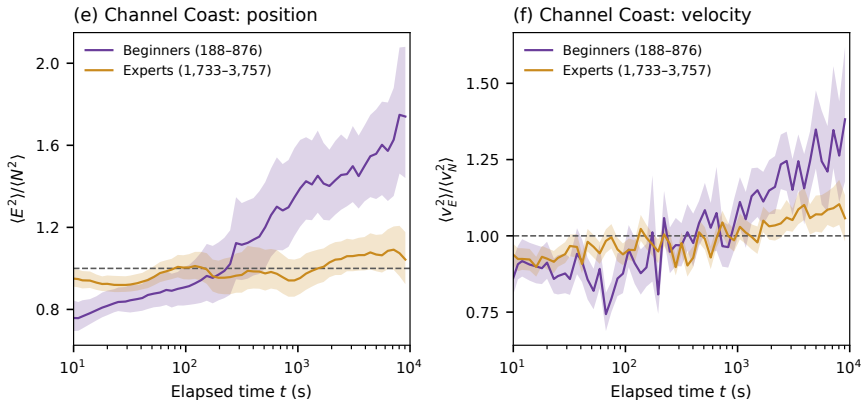
Alps: C/D/CCC has smaller coordinate-moment imbalance



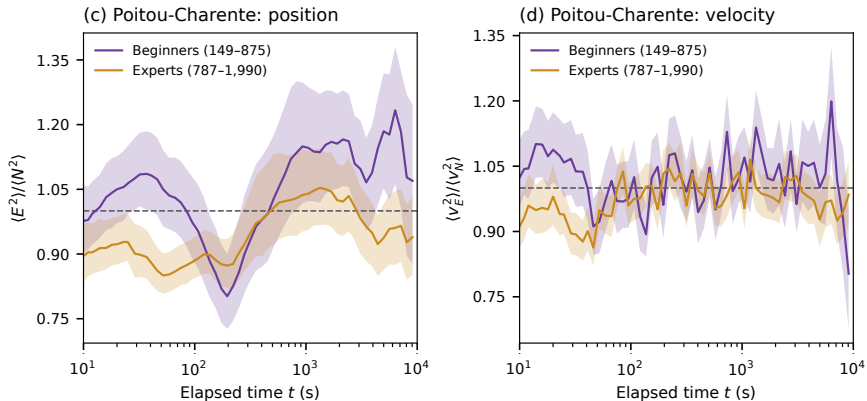
Pyrenees: the ordering of the two groups reverses

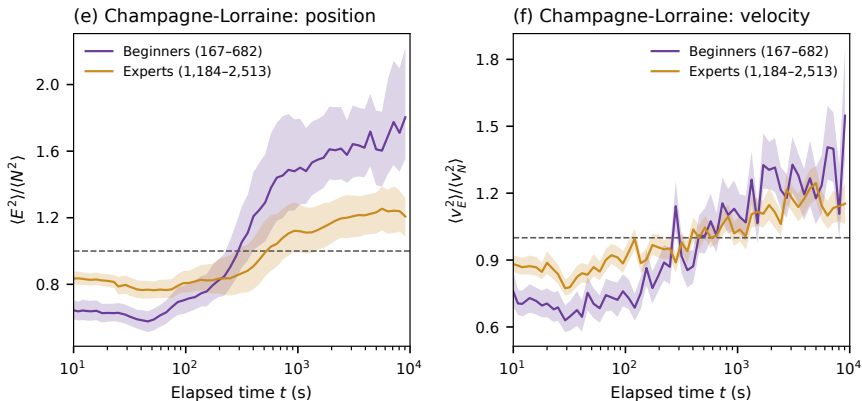


Channel Coast: the late position contrast is pronounced



Poitou-Charente: the group curves cross





Hypothesis to discuss

Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

Evidence limiting that interpretation

The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

$$\mathbf{V}_{\text{ground}} = \mathbf{V}_{\text{air}} + \mathbf{W}_h.$$

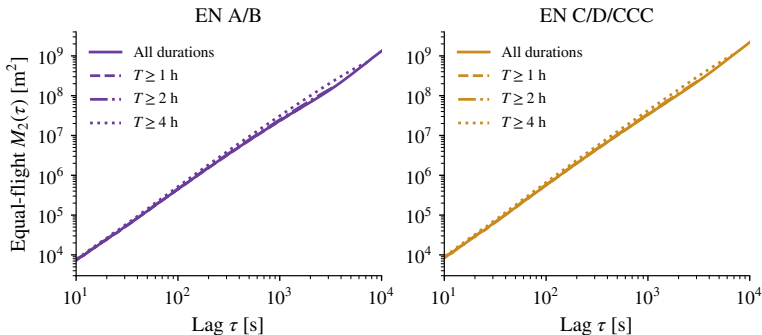
Upwind progress, air-relative performance, choice of route and geographic moment balance are different outcomes.

A first comparison would match overlapping site-day, duration and task support; then separate mean drift, centred variability and individual route orientation. Regional references show terrain alignment and a remaining coastal wind-PCA offset at flight altitude; local track-matched observations are still needed.

Thesis §3.6

Flight duration and equipment composition

Long flights occur in both equipment groups



For $T \geq 4$ h: **7,415 A/B flights** and **27,815 C/D/CCC flights**.

Across 10–10,000 s, all-duration slopes are **1.728** and **1.788**.

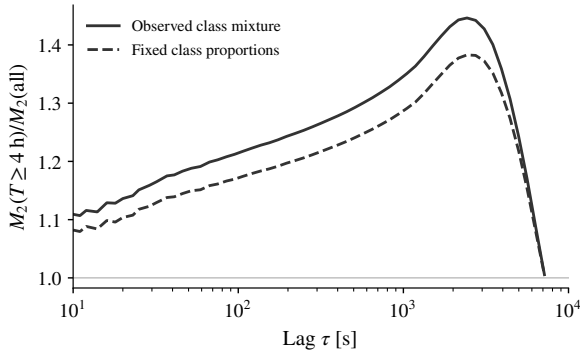
Equal-flight MSD; groups are nested. Threshold curves stop at $\tau = T_0/2$; at least 30 flights per lag. [All counts and common-range slopes](#).

Duration and equipment on the common 10–1695 s interval

Experience proxy	Duration	Flights	b	RMS residual (dex)
Beginners	All	54,292	1.755	0.022
Beginners	$T \geq 1$ h	51,124	1.754	0.022
Beginners	$T \geq 2$ h	31,768	1.767	0.020
Beginners	$T \geq 4$ h	7,415	1.806	0.017
Experts	All	94,283	1.803	0.017
Experts	$T \geq 1$ h	91,182	1.802	0.017
Experts	$T \geq 2$ h	69,087	1.811	0.016
Experts	$T \geq 4$ h	27,815	1.836	0.015

Counts are eligible flights, not independent windows; support falls with lag. RMS residual measures departures from a fitted line, not uncertainty. Threshold samples overlap.

Fixing equipment proportions removes part of the duration contrast



$$M_{T_0}^{\text{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau).$$

Fixed weights: $\pi_{A/B} = 0.365$,
 $\pi_{C/D/CCC} = 0.635$.

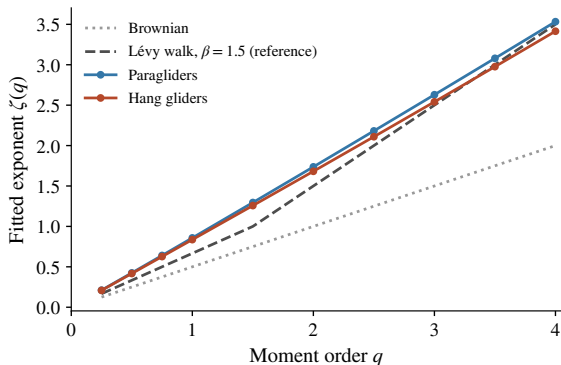
At $\tau = 1695$ s the ratio falls from 1.42 to 1.35. A within-class contrast remains.

Both curves use the same two known equipment groups. Equality at one lag does not establish independence from flight duration.

Thesis §3.7.1

The moment spectrum and the Lévy-walk benchmark

The moment spectrum challenges a specific Lévy-walk benchmark



$$M_q = \langle R^q \rangle \propto \tau^{\zeta(q)}.$$

For a constant-speed renewal walk with isotropic independent headings and $\psi(t) \sim t^{-1-\beta}$, $1 < \beta < 2$:

$$\zeta_{\text{LW}}(q) = \begin{cases} q/\beta, & q < \beta, \\ q + 1 - \beta, & q > \beta. \end{cases}$$

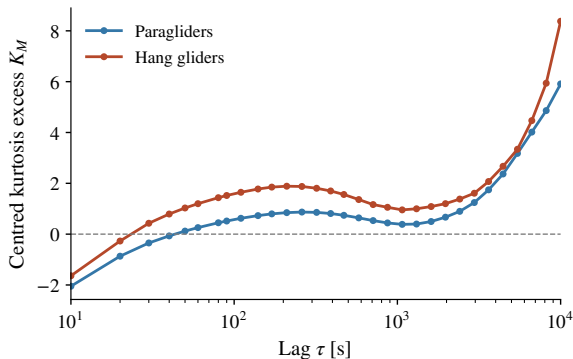
The empirical spectrum is nearly linear over $q = 0.25$ –4.

Pooled windows. $\beta = 1.5$ is illustrative, not a fitted null; $q = \beta$ has a logarithmic correction. [Zaburdaev, Denisov & Klafter \(2015\)](#).

Thesis §3.7.2

Centred kurtosis and heterogeneity

Non-Gaussian pooled increments can arise from Gaussian mixtures



$$\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu}),$$

$$K_M = \langle |\mathbf{Z}|^4 \rangle - 8.$$

A nonsingular 2D Gaussian has $K_M = 0$, including anisotropic Gaussians.

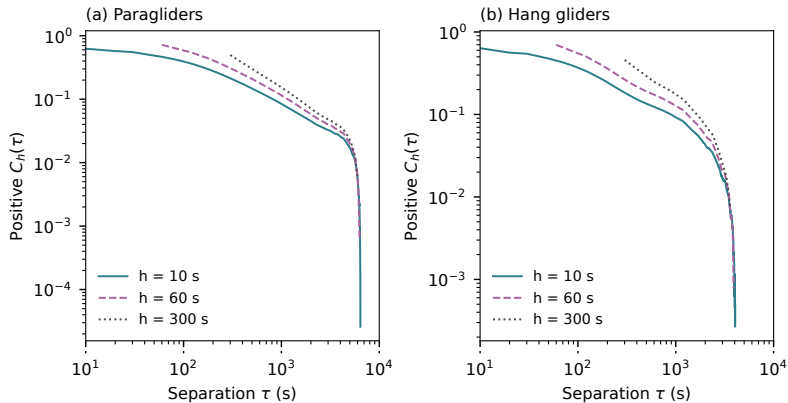
The observed curve changes sign and rises at large lag, where support is weakest.

Centred, full-covariance Mardia excess; pooled windows. Positive excess does not identify a power-law tail. [Mardia \(1970\)](#).

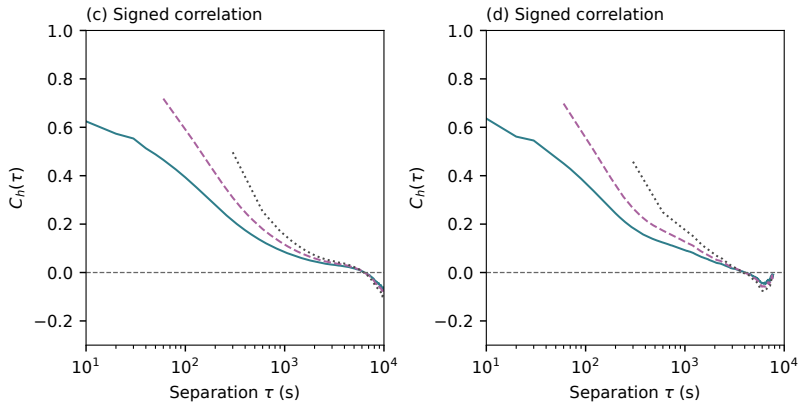
Thesis §3.7.3

Velocity memory across resolutions

Velocity memory remains visible after averaging over minutes



Signed correlations retain the long-lag negative values



For the same stationary, centred velocity population,

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s)C(s) ds.$$

With $C(s) = e^{-s/t_p}$,

$$M_{2,c} = 2\sigma_v^2 t_p [\tau - t_p(1 - e^{-\tau/t_p})].$$

Ballistic at short times, diffusive at long times.

[Taylor \(1922\)](#), pp. 207–208. This covariance identity cannot be applied directly to differently weighted heterogeneous means.

Thesis §3.7.4

Model predictions and remaining tests

Current constraints on candidate models

Candidate	Prediction to test	Present conclusion
Homogeneous Brownian	Linear MSD; independent increments	Challenged over the measured range.
Constant drift + Brownian	Drift-free $V_2 \propto \tau$	Simple version challenged; mixtures unresolved.
Homogeneous fBm	Gaussian law; common H_p	Challenged as one pooled law.
Exponential velocity memory	Ballistic–diffusive crossover	Remains a candidate.
Renewal Lévy walk, $1 < \beta < 2$	Two-branch moment spectrum	Finite-record comparison still needed.

“Challenged” means a descriptive mismatch. Calibrated statistical rejection is still to be done.

Each candidate needs its own parameters, environmental assumptions and observation protocol. A long-time diffusive limit has not been tested.

Thesis §3.8

Constraints for the phase analysis

$$M_q(\tau) = \sum_c w_c(\tau) M_q(\tau | c), \quad w_c = \frac{n_c}{\sum_{c'} n_{c'}}.$$

Here c partitions all admissible windows, including phase histories that cross boundaries. With equal flight weights, apply the identity within each flight before averaging.

The identity holds for any exhaustive labels. Recovering the global curve verifies accounting, not the physical accuracy of those labels.

Three decisions for this meeting

1. Define the model target.

Ground-relative motion conditional on environment; residual motion requires an independently justified wind correction.

2. Agree the first calibrated comparisons.

Fit Brownian controls, fBm, finite persistence and a specified Lévy walk; simulate the same record lengths, selection and estimators.

3. Choose the most informative next measurements.

Site-day uncertainty and route matching now; phase durations, increments and between-phase dependence after segmentation validation.

We have measured constraints on growth, shape, geometry and memory. The generating process and an intrinsic Hurst exponent remain unresolved.

$$Q_{Y^2}(p) = Q_Y(p)^2, \quad H_{Y^2,p} = 2H_{Y,p} \quad \text{for } Y \geq 0.$$

Squaring bin edges preserves their probability masses. A monotone transform also preserves the supremum distance between the correspondingly transformed CDFs.

If $f_Y(0^+) = c > 0$, then $f_{Y^2}(s) \sim c/(2\sqrt{s})$ near zero. This Jacobian effect says nothing about a heavy upper tail.

Zaburdaev, Denisov & Klafter (2015), *Lévy walks*

Moment-spectrum benchmark

Mandelbrot & Van Ness (1968), *Fractional Brownian motions*

fBm construction

Taylor (1922), *Diffusion by continuous movements*

Velocity/displacement relation

Mardia (1970), *Measures of multivariate skewness and kurtosis*

Centred kurtosis

Efron (1979), *Bootstrap methods*

Resampling principle

Numerical evidence: current Chapter 3 reports, frozen in the accompanying data and asset manifests. Bibliographic access limitations are retained in the repository audit; citations do not validate the chosen sampling design or constitute calibrated rejection tests.

Fraiman et al., Cramér–Wold results

Didier & Pipiras, operator fractional Brownian motions

DHV classification guidance

DREAL Normandie, Le Pays de Caux

Joint laws and projections

Beyond scalar rescaling

Class is not a performance measure

Coastal cliffs and valleys

The cited pages support specific mathematical or observational cautions. They do not identify the causes of the regional contrasts measured here.